

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: ['r_temperature_fft_8h_fft_f3_cph', 'r_temperature_fft_8h_fft_period3_h', 'r_temperature_fft_8h_fft_power3']

1.2 Metadata

Nombre de lignes df	7072
Nombre de lignes df clean	7072
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	4950
Taille dataset test	2121
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_8h_fft_f1_cph, r_NH3_fft_8h_fft_period1_h, r_NH3_fft_8h_fft_power1, r_NH3_fft_8h_fft_f2_cph, r_NH3_fft_8h_fft_period2_h, r_NH3_fft_8h_fft_power2, r_NH3_fft_8h_fft_f3_cph, r_NH3_fft_8h_fft_period3_h, r_NH3_fft_8h_fft_power3, r_temperature_fft_8h_fft_f1_cph, r_temperature_fft_8h_fft_period1_h, r_temperature_fft_8h_fft_power1, r_temperature_fft_8h_fft_f2_cph, r_temperature_fft_8h_fft_period2_h, r_temperature_fft_8h_fft_power2, r_umidity_fft_8h_fft_f1_cph, r_umidity_fft_8h_fft_period1_h, r_umidity_fft_8h_fft_power1, r_umidity_fft_8h_fft_f2_cph, r_umidity_fft_8h_fft_period2_h, r_umidity_fft_8h_fft_power2, r_umidity_fft_8h_fft_f3_cph, r_umidity_fft_8h_fft_period3_h, r_umidity_fft_8h_fft_power3

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

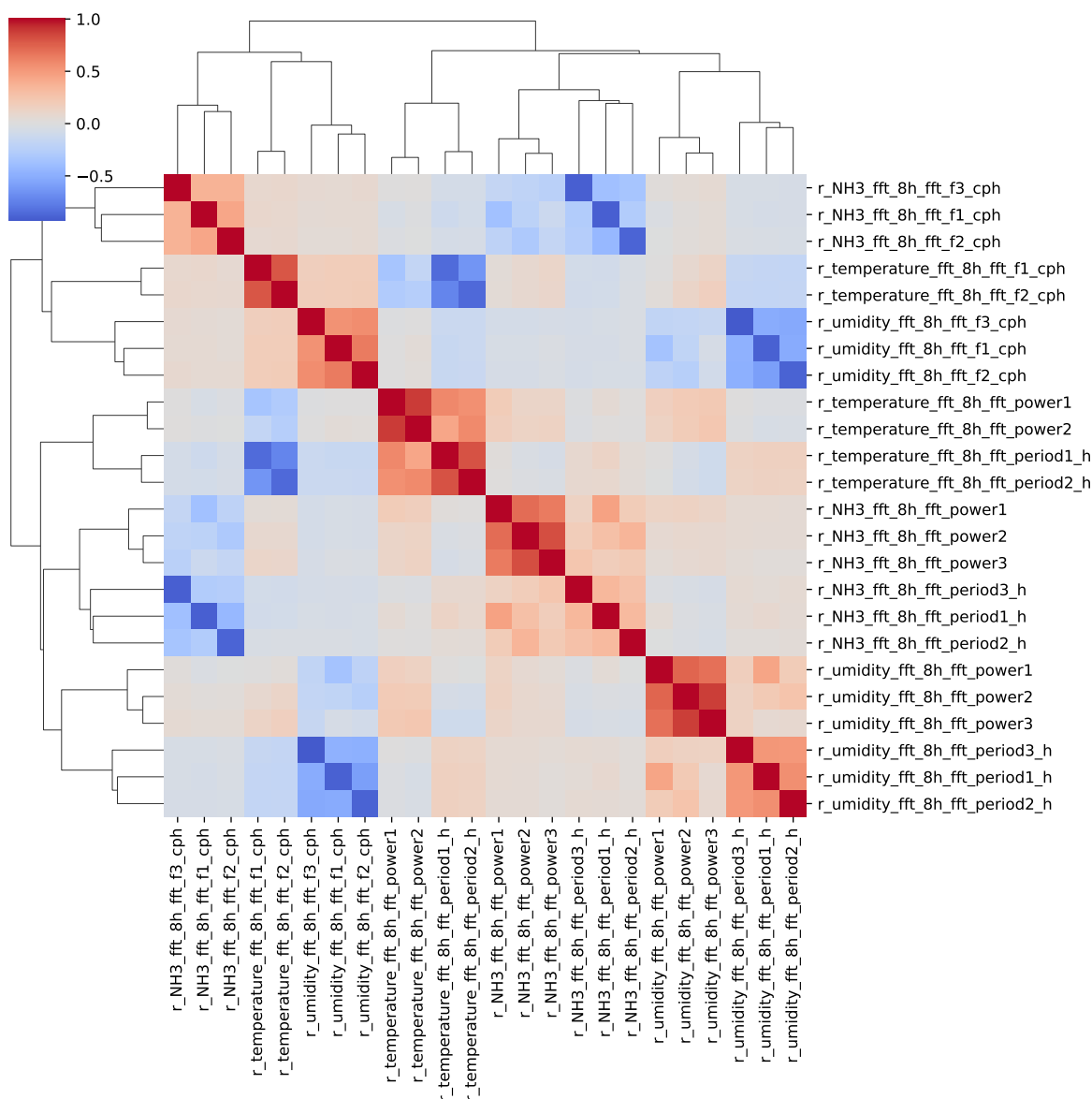
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_fft_8h_fft_period1_h, r_NH3_fft_8h_fft_f1_cph ; **cluster 7:** r_NH3_fft_8h_fft_power1 ; **cluster 8:** r_NH3_fft_8h_fft_f2_cph, r_NH3_fft_8h_fft_period2_h ; **cluster 9:** r_NH3_fft_8h_fft_power2, r_NH3_fft_8h_fft_power3 ; **cluster 10:** r_NH3_fft_8h_fft_period3_h, r_NH3_fft_8h_fft_f3_cph ; **cluster 11:** r_temperature_fft_8h_fft_f2_cph, r_temperature_fft_8h_fft_f1_cph, r_temperature_fft_8h_fft_period2_h, r_temperature_fft_8h_fft_period1_h ; **cluster 12:** r_temperature_fft_8h_fft_power2, r_temperature_fft_8h_fft_power1 ; **cluster 13:** r_umidity_fft_8h_fft_period1_h, r_umidity_fft_8h_fft_f1_cph ; **cluster 14:** r_umidity_fft_8h_fft_power1 ; **cluster 15:** r_umidity_fft_8h_fft_period2_h, r_umidity_fft_8h_fft_f2_cph ; **cluster 16:** r_umidity_fft_8h_fft_power2, r_umidity_fft_8h_fft_power3 ; **cluster 17:** r_umidity_fft_8h_fft_period3_h, r_umidity_fft_8h_fft_f3_cph ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_8h_fft_period1_h, r_NH3_fft_8h_fft_power1, r_NH3_fft_8h_fft_f2_cph, r_NH3_fft_8h_fft_power2, r_NH3_fft_8h_fft_period3_h, r_temperature_fft_8h_fft_f2_cph, r_temperature_fft_8h_fft_power2, r_umidity_fft_8h_fft_period1_h, r_umidity_fft_8h_fft_power1, r_umidity_fft_8h_fft_period2_h, r_umidity_fft_8h_fft_power2, r_umidity_fft_8h_fft_period3_h



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_8h_fft_period1_h, r_NH3_fft_8h_fft_power1, r_NH3_fft_8h_fft_f2_cph, r_NH3_fft_8h_fft_power2, r_NH3_fft_8h_fft_period3_h, r_temperature_fft_8h_fft_f2_cph, r_temperature_fft_8h_fft_power2,

r_umidity_fft_8h_fft_period1_h, r_umidity_fft_8h_fft_power1, r_umidity_fft_8h_fft_period2_h, r_umidity_fft_8h_fft_power2, r_umidity_fft_8h_fft_period3_h

Len(Features_X): 17

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 4950 samples Test: 2122 samples



3.3.2 Internal CV Folds

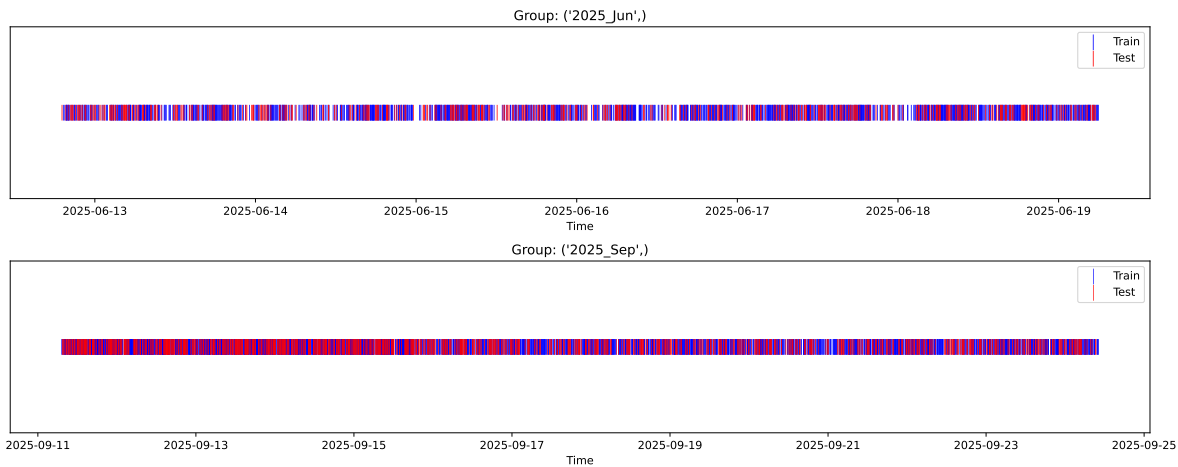
Fold 1/5

Train: 3465 samples Test: 1485 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3465 samples Test: 1485 samples Excluded: 0 samples Window exclusion: 0h



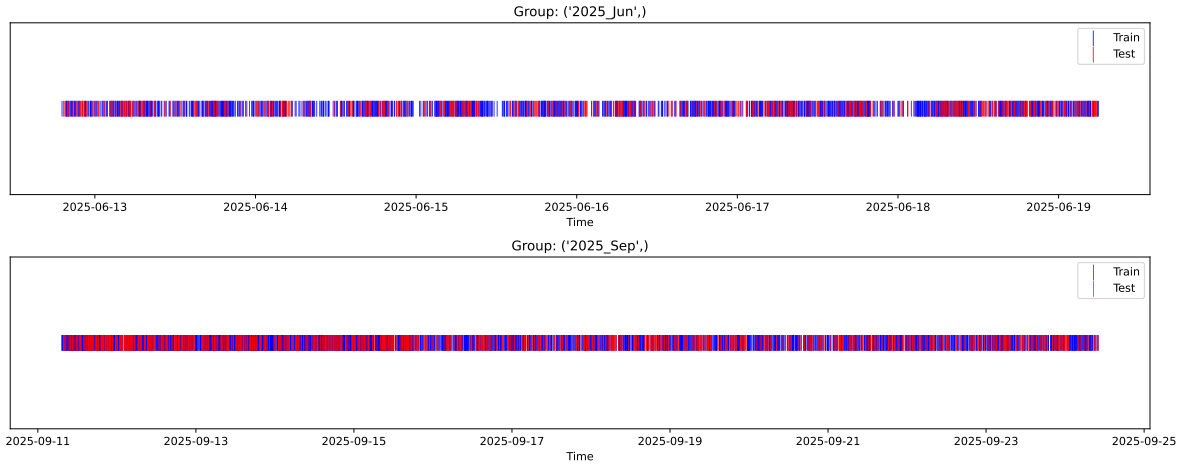
Fold 3/5

Train: 3465 samples Test: 1485 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3465 samples Test: 1485 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3465 samples Test: 1485 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.336, std=0.788, min=0.959, max=8.682
 y_{test} : mean=2.345, std=0.834, min=1.062, max=13.845

Fold 0 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.33, std=0.79

Fold 1 — y_{test} : mean=2.32, std=0.75, y_{train} : mean=2.34, std=0.80

Fold 2 — y_{test} : mean=2.36, std=0.80, y_{train} : mean=2.33, std=0.78

Fold 3 — y_test: mean=2.30, std=0.76, y_train: mean=2.35, std=0.80

Fold 4 — y_test: mean=2.35, std=0.79, y_train: mean=2.33, std=0.79

train scores CV: [0.39565599 0.38892516 0.39383142 0.39769458 0.39001537]

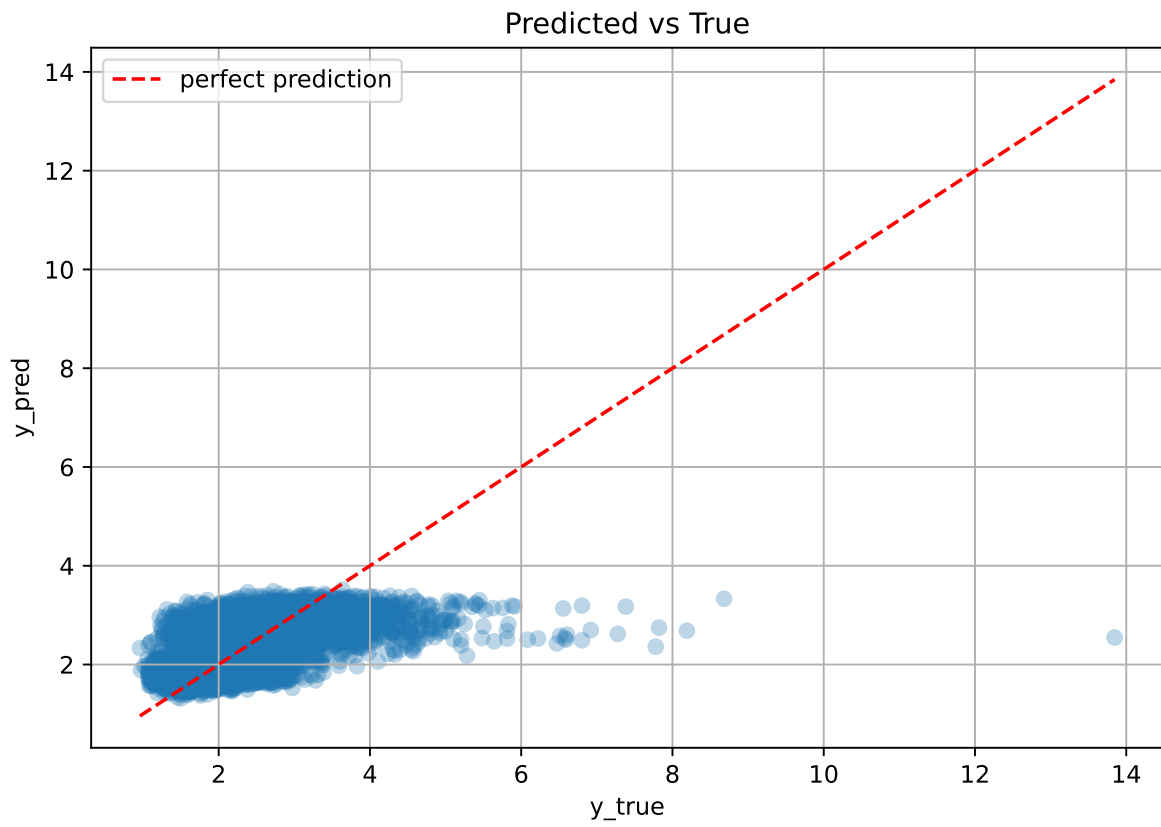
test scores CV: [0.38101597 0.39621743 0.38519885 0.37218506 0.39364984]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3932	0.3857	0.0087	0.3365	0.3377	-0.0479	0.8758	1.0196

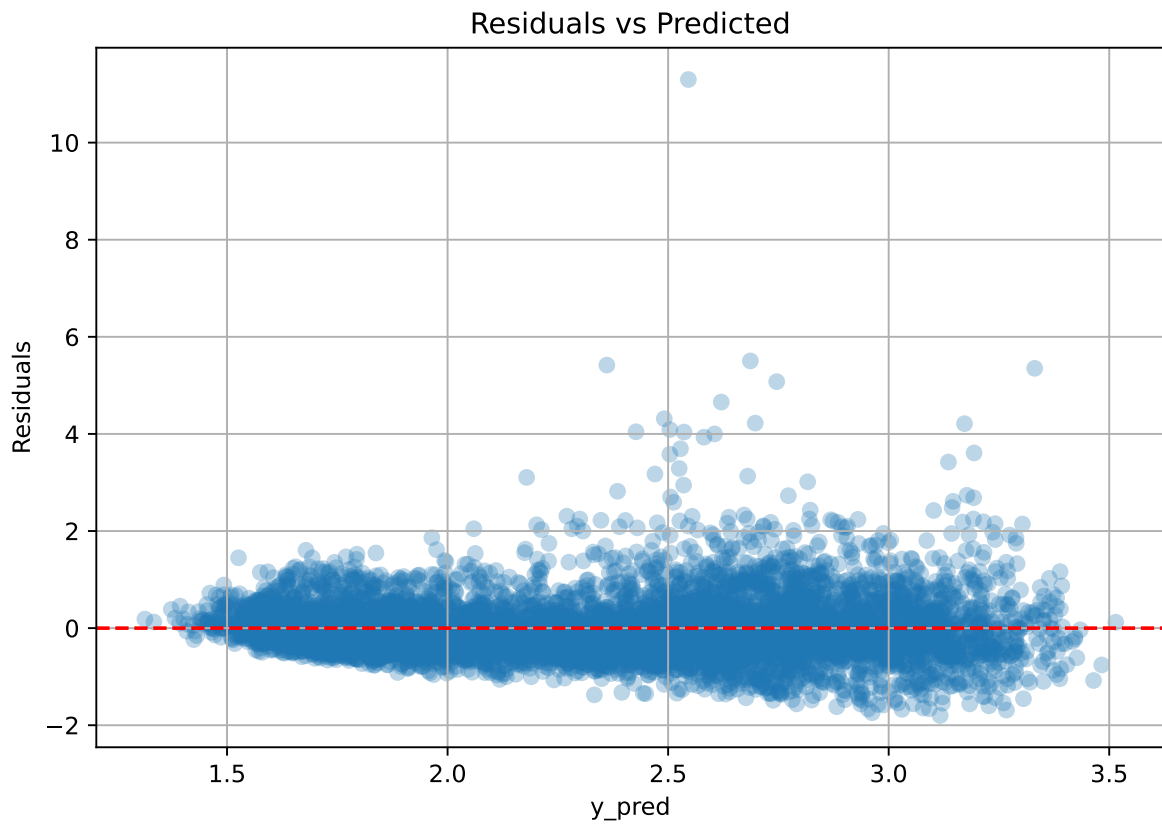
Coefficients:

coef_r_CO2 = -0.0381, coef_r_NH3 = -0.0489, coef_r_NH3_fft_8h_fft_f2_cph = -0.0419, coef_r_NH3_fft_8h_fft_period1_h = -0.0026, coef_r_NH3_fft_8h_fft_period3_h = 0.0202, coef_r_NH3_fft_8h_fft_power1 = 0.0290, coef_r_NH3_fft_8h_fft_power2 = -0.0097, coef_r_THI = 0.4337, coef_r_temperature = -0.4944, coef_r_temperature_fft_8h_fft_f2_cph = -0.0056, coef_r_temperature_fft_8h_fft_power2 = -0.0134, coef_r_umidity = -0.6055, coef_r_umidity_fft_8h_fft_period1_h = 0.0322, coef_r_umidity_fft_8h_fft_period2_h = -0.0014, coef_r_umidity_fft_8h_fft_period3_h = -0.0286, coef_r_umidity_fft_8h_fft_power1 = -0.0559, coef_r_umidity_fft_8h_fft_power2 = -0.0291, intercept = 2.3360,

3.4.2 Scatter



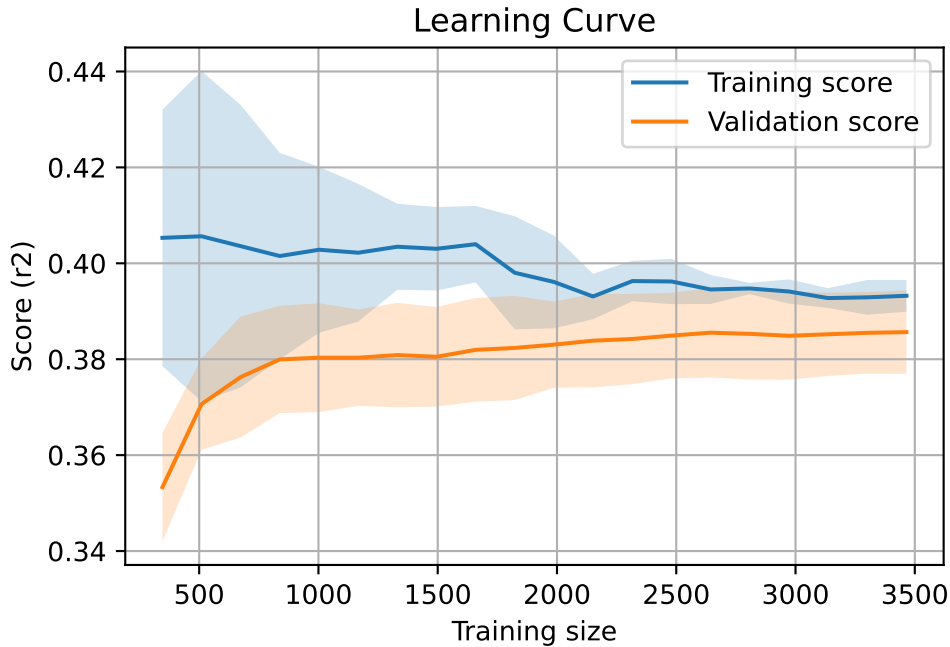
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7072, `Len(training)` : 4951,

`Len(training_real)` : 3466, `Len(test_of_training)` : 1485, `Len(test)` : 2121



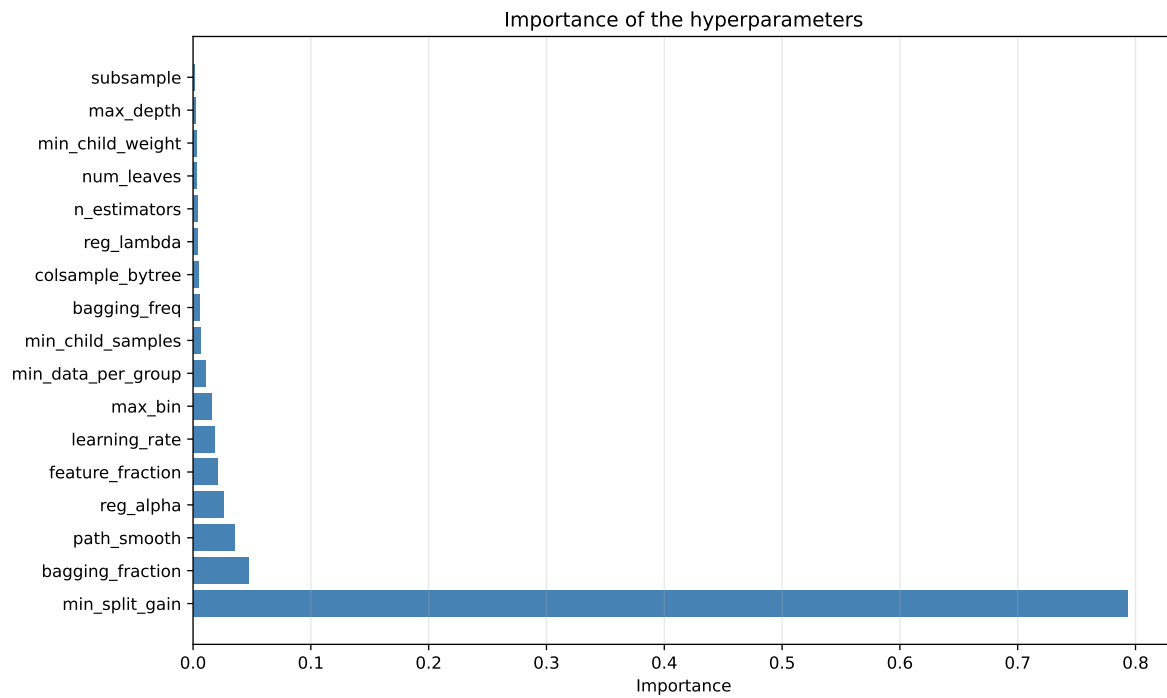
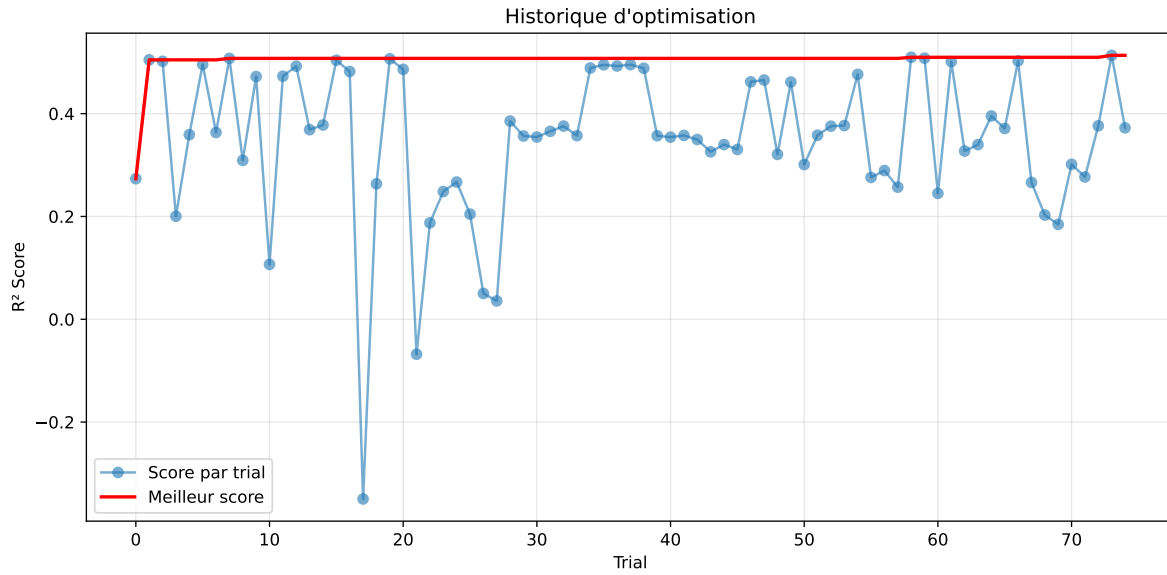
3.5 Model : LGBM

3.5.1 Gridsearch

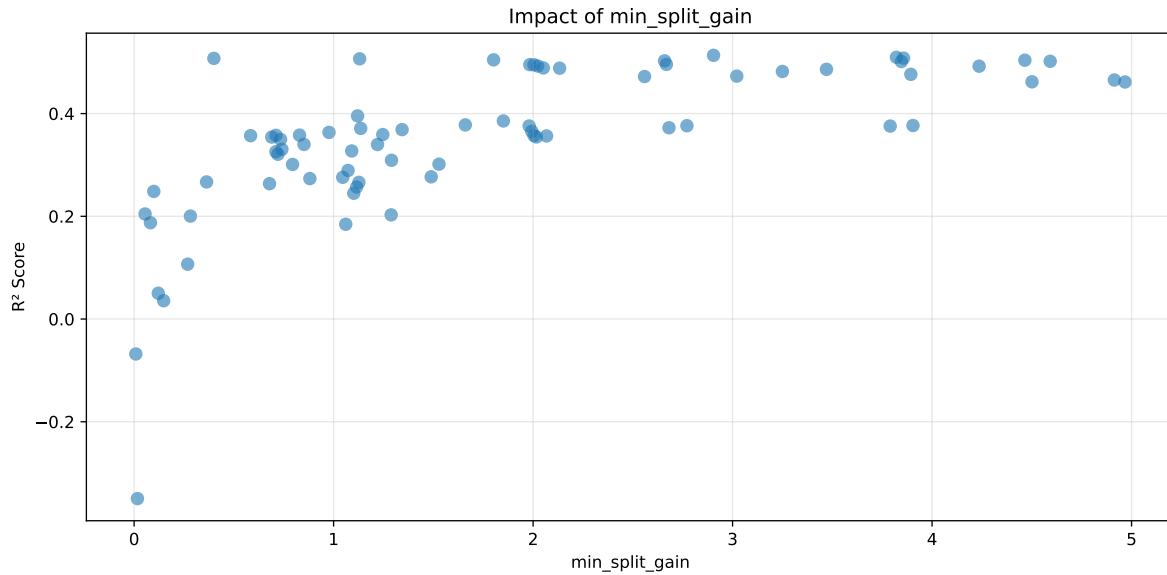
Distribution y_train vs y_test: y_train: mean=2.336, std=0.788, min=0.959, max=8.682
y_test: mean=2.345, std=0.834, min=1.062, max=13.845

===== TOP Importance FEATURES =====

feature	importance
r_umidity	631
r_CO2	605
r_NH3	398
r_THI	316
r_temperature	264
r_NH3_fft_8h_fft_power1	206
r_temperature_fft_8h_fft_power2	161
r_NH3_fft_8h_fft_power2	128
r_umidity_fft_8h_fft_power2	119
r_NH3_fft_8h_fft_f2_cph	87
r_temperature_fft_8h_fft_f2_cph	84
r_NH3_fft_8h_fft_period1_h	69
r_umidity_fft_8h_fft_power1	37
r_umidity_fft_8h_fft_period1_h	26
r_NH3_fft_8h_fft_period3_h	10
r_umidity_fft_8h_fft_period2_h	9
r_umidity_fft_8h_fft_period3_h	7



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5378	0.5134	0.4445	0.0087	0.5134	0.4684	-0.045	0.9123	1.0475
2	0.5351	0.5096	0.4421	0.0103	0.5096	0.466	-0.0436	0.9144	1.05
3	0.5334	0.508	0.4404	0.0097	0.508	0.4666	-0.0414	0.9186	1.0499
4	0.5286	0.5075	0.4363	0.0087	0.5075	0.4505	-0.057	0.8877	1.0415
5	0.5311	0.5066	0.4403	0.0113	0.5066	0.4639	-0.0427	0.9157	1.0483
6	0.5241	0.5047	0.4328	0.0092	0.5047	0.4547	-0.05	0.9009	1.0384
7	0.5217	0.5039	0.4297	0.0109	0.5039	0.4533	-0.0506	0.8995	1.0353
8	0.5238	0.5027	0.4324	0.0093	0.5027	0.4521	-0.0506	0.8993	1.0421
9	0.5251	0.5018	0.4361	0.0115	0.5018	0.4624	-0.0394	0.9214	1.0464
10	0.5257	0.5013	0.4354	0.0092	0.5013	0.4629	-0.0384	0.9234	1.0487
---	---	---	---	---	---	---	---	---	---
75	0.8541	0.6535	0.5755	0.0092	-0.3496	0.598	-0.0555	0.9151	1.307

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.010265003391001278, 'reg__num_leaves': 48, 'reg__subsample': 0.6945108742253695, 'reg__colsample_bytree': 0.886447065793767, 'reg__min_child_samples': 13, 'reg__reg_alpha': 0.3089169331335112, 'reg__reg_lambda': 26.658783152133545, 'reg__min_split_gain': 2.9045254335872155, 'reg__path_smooth': 4.335060818426918, 'reg__min_child_weight': 0.016595628178631815, 'reg__feature_fraction': 0.6421237980035186, 'reg__bagging_fraction': 0.5459732517496424, 'reg__bagging_freq': 5, 'reg__max_bin': 190, 'reg__min_data_per_group': 180}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.011447083677126251, 'reg__num_leaves': 56, 'reg__subsample': 0.6513564714252812, 'reg__colsample_bytree': 0.8821349720728611, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.9780877596138691, 'reg__reg_lambda': 23.797727531538325, 'reg__min_split_gain': 3.8213686231630817, 'reg__path_smooth': 2.2934317960889556, 'reg__min_child_weight': 0.07008371249412276, 'reg__feature_fraction': 0.6159503595204123, 'reg__bagging_fraction': 0.8067852356969422, 'reg__bagging_freq': 7, 'reg__max_bin': 236, 'reg__min_data_per_group': 169}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.011228611679343586, 'reg__num_leaves': 58, 'reg__subsample': 0.670645425675769, 'reg__colsample_bytree': 0.7298785095853468, 'reg__min_child_samples': 23, 'reg__reg_alpha': 1.294320027537294, 'reg__reg_lambda': 24.10026184173923, 'reg__min_split_gain': 3.8571368012634744, 'reg__path_smooth': 2.3354948940628075, 'reg__min_child_weight': 0.020205389775465276, 'reg__feature_fraction': 0.608812935165268, 'reg__bagging_fraction': 0.8150262965532451, 'reg__bagging_freq': 3, 'reg__max_bin': 171, 'reg__min_data_per_group': 172}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.01847408192053824, 'reg__num_leaves': 24, 'reg__subsample': 0.6985888420975634, 'reg__colsample_bytree': 0.53814806034854, 'reg__min_child_samples': 43, 'reg__reg_alpha': 3.8422312645587575, 'reg__reg_lambda': 1.6151114951787116, 'reg__min_split_gain': 0.39973410668882436, 'reg__path_smooth': 1.5670454763827124, 'reg__min_child_weight': 1.6816893764867373, 'reg__feature_fraction': 0.5763750987601777, 'reg__bagging_fraction': 0.5473255844924828, 'reg__bagging_freq': 3, 'reg__max_bin': 83, 'reg__min_data_per_group': 52}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.010824587321623835, 'reg__num_leaves': 61, 'reg__subsample': 0.783068488591812, 'reg__colsample_bytree': 0.6795799527947208, 'reg__min_child_samples': 45, 'reg__reg_alpha': 7.522174214826128, 'reg__reg_lambda': 18.90413508975391, 'reg__min_split_gain': 1.1304631796595448, 'reg__path_smooth': 4.409771318212932, 'reg__min_child_weight': 0.15719468463447145, 'reg__feature_fraction': 0.8546902187018648, 'reg__bagging_fraction': 0.6055347570928853, 'reg__bagging_freq': 2, 'reg__max_bin': 95, 'reg__min_data_per_group': 118}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.09998836329945036, 'reg__num_leaves': 23, 'reg__subsample': 0.5592889141994961, 'reg__colsample_bytree': 0.622503307848586, 'reg__min_child_samples': 24, 'reg__reg_alpha': 9.17483631621133, 'reg__reg_lambda': 44.18458703871032, 'reg__min_split_gain': 1.8019094148552512, 'reg__path_smooth': 3.2236908522554013, 'reg__min_child_weight': 1.3786359724986799, 'reg__feature_fraction': 0.7634421991070535, 'reg__bagging_fraction': 0.7618087923499653, 'reg__bagging_freq': 4, 'reg__max_bin': 253, 'reg__min_data_per_group': 106}

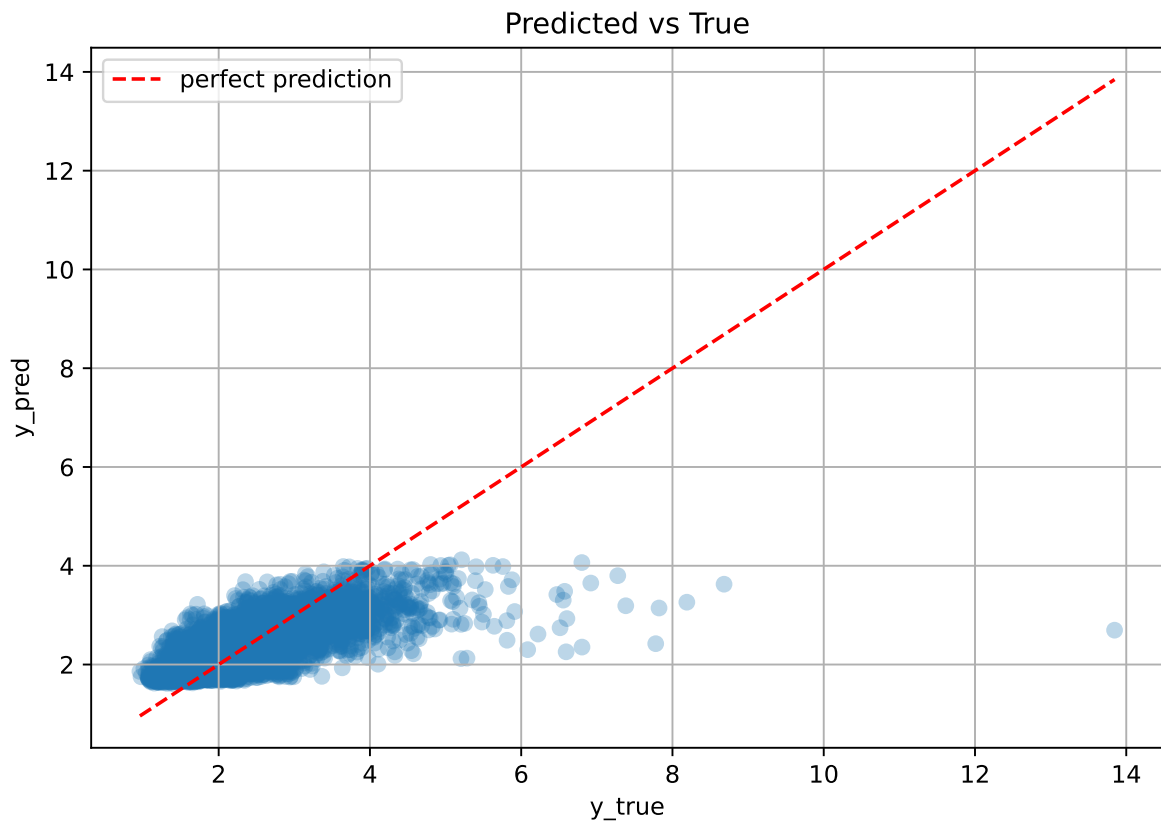
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.08146128070840032, 'reg__num_leaves': 16, 'reg__subsample': 0.7684353890752036, 'reg__colsample_bytree': 0.5491926435887403, 'reg__min_child_samples': 22, 'reg__reg_alpha': 0.38932600238317017, 'reg__reg_lambda': 2.230765142720709, 'reg__min_split_gain': 4.465163878999741, 'reg__path_smooth': 4.711967774047075, 'reg__min_child_weight': 0.2558596619774843, 'reg__feature_fraction': 0.6242516423872917, 'reg__bagging_fraction': 0.5829720736545225, 'reg__bagging_freq': 7, 'reg__max_bin': 82, 'reg__min_data_per_group': 153}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.02098569980040333, 'reg__num_leaves': 20, 'reg__subsample': 0.6980101969174547, 'reg__colsample_bytree': 0.8993561462333269, 'reg__min_child_samples': 47, 'reg__reg_alpha': 5.688067629080542, 'reg__reg_lambda': 1.5375859344141178, 'reg__min_split_gain': 2.658984900797278, 'reg__path_smooth': 2.7539505340430233, 'reg__min_child_weight': 0.020565843233617393, 'reg__feature_fraction': 0.525581983734198, 'reg__bagging_fraction': 0.8230518818356778, 'reg__bagging_freq': 7, 'reg__max_bin': 171, 'reg__min_data_per_group': 151}

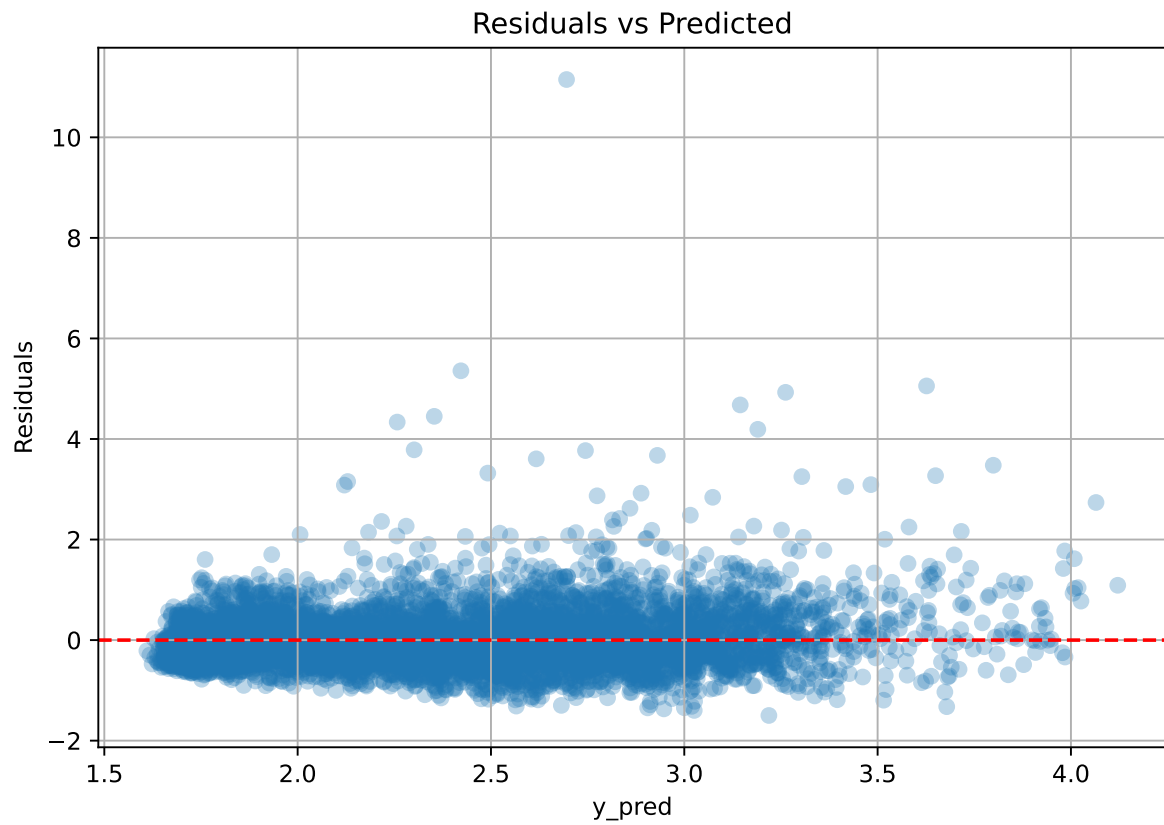
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.010054405522510445, 'reg__num_leaves': 59, 'reg__subsample': 0.6573839219280689, 'reg__colsample_bytree': 0.8515712142309719, 'reg__min_child_samples': 24, 'reg__reg_alpha': 1.0392611686915343, 'reg__reg_lambda': 21.14269883099841, 'reg__min_split_gain': 4.592186712312441, 'reg__path_smooth': 2.389646091520757, 'reg__min_child_weight': 0.058650162212913304, 'reg__feature_fraction': 0.6120722859069379, 'reg__bagging_fraction': 0.8483576618382567, 'reg__bagging_freq': 2, 'reg__max_bin': 223, 'reg__min_data_per_group': 155}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.01164266254373295, 'reg__num_leaves': 59, 'reg__subsample': 0.6607450810599725, 'reg__colsample_bytree': 0.74104910759728, 'reg__min_child_samples': 45, 'reg__reg_alpha': 1.1286028999251654, 'reg__reg_lambda': 23.77599245903347, 'reg__min_split_gain': 3.846075444216596, 'reg__path_smooth': 2.7384654255817815, 'reg__min_child_weight': 0.06753843184634005, 'reg__feature_fraction': 0.6030938544311263, 'reg__bagging_fraction': 0.8125372849929771, 'reg__bagging_freq': 7, 'reg__max_bin': 171, 'reg__min_data_per_group': 169}

3.5.2 Scatter



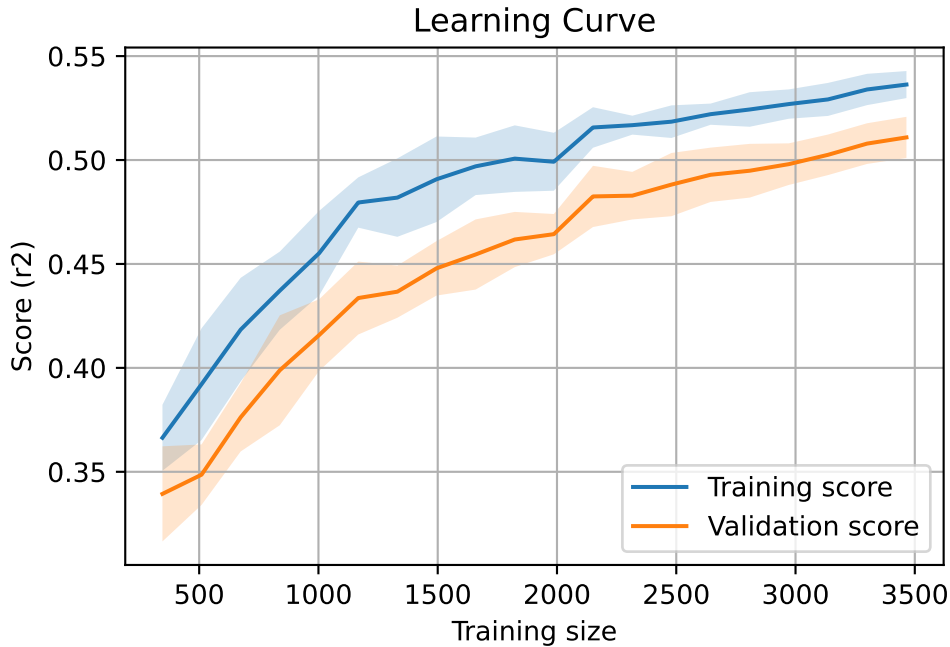
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7072, `Len(training)` : 4951,

`Len(training_real)` : 3466, `Len(test_of_training)` : 1485, `Len(test)` : 2121

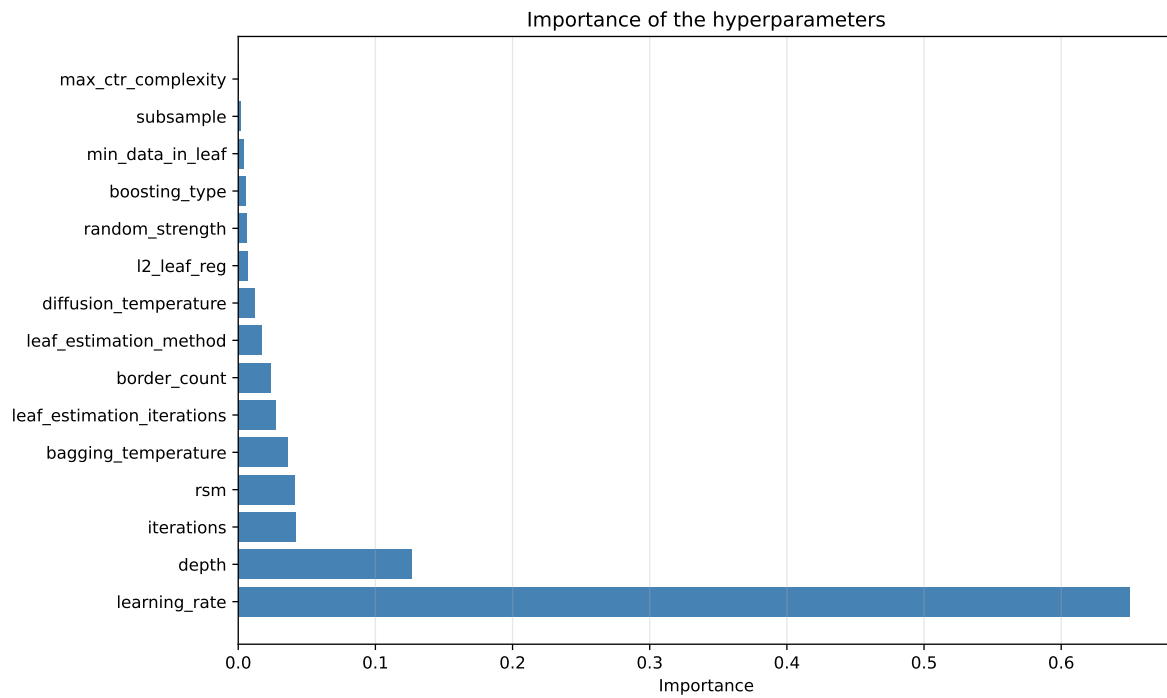
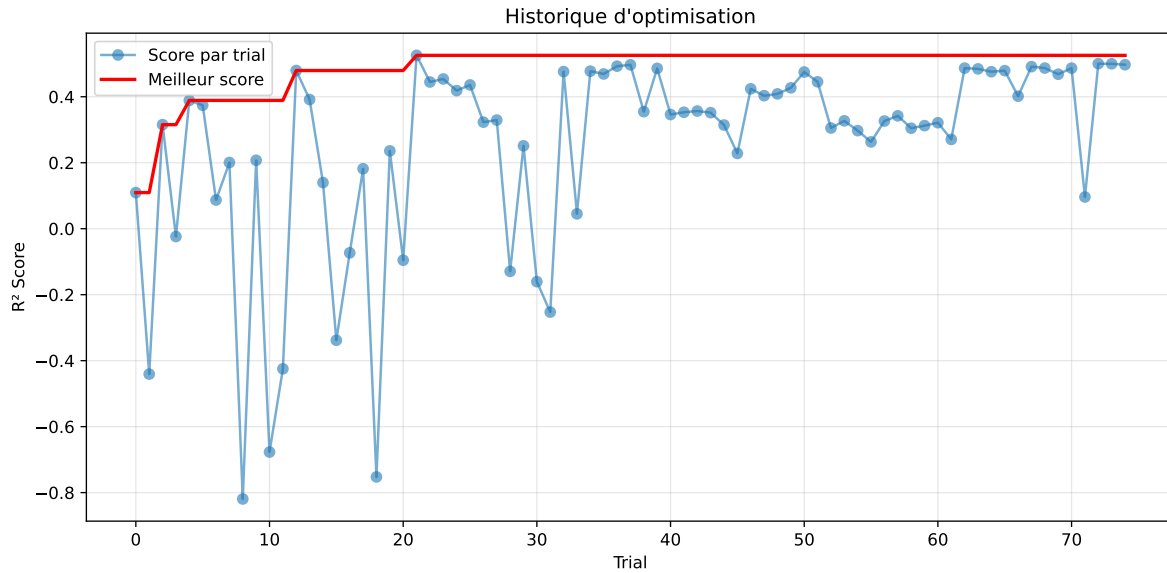


3.6 Model : CatBoost

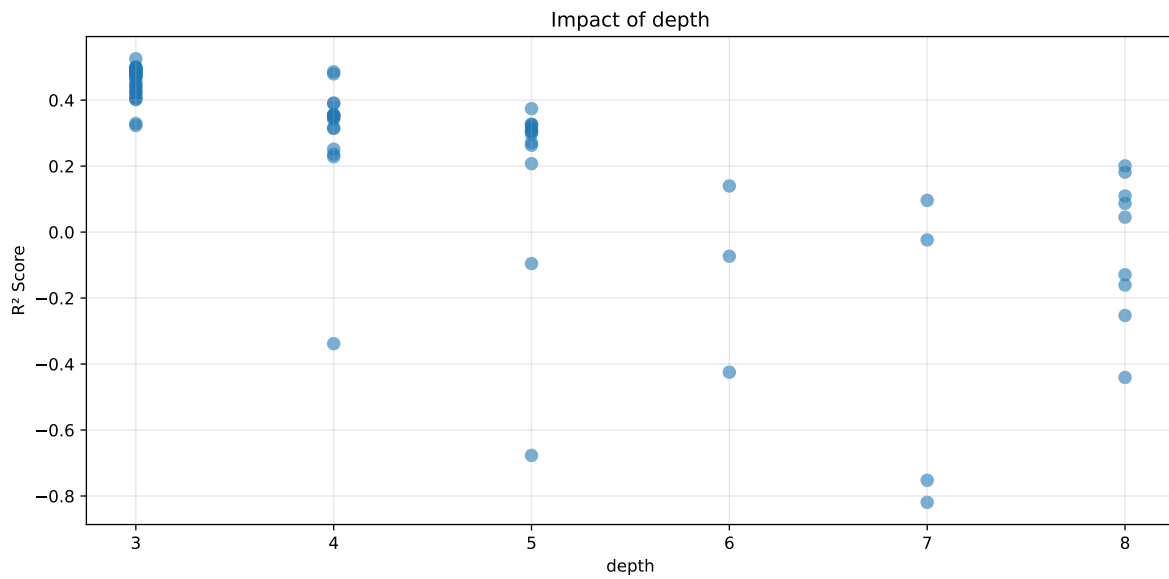
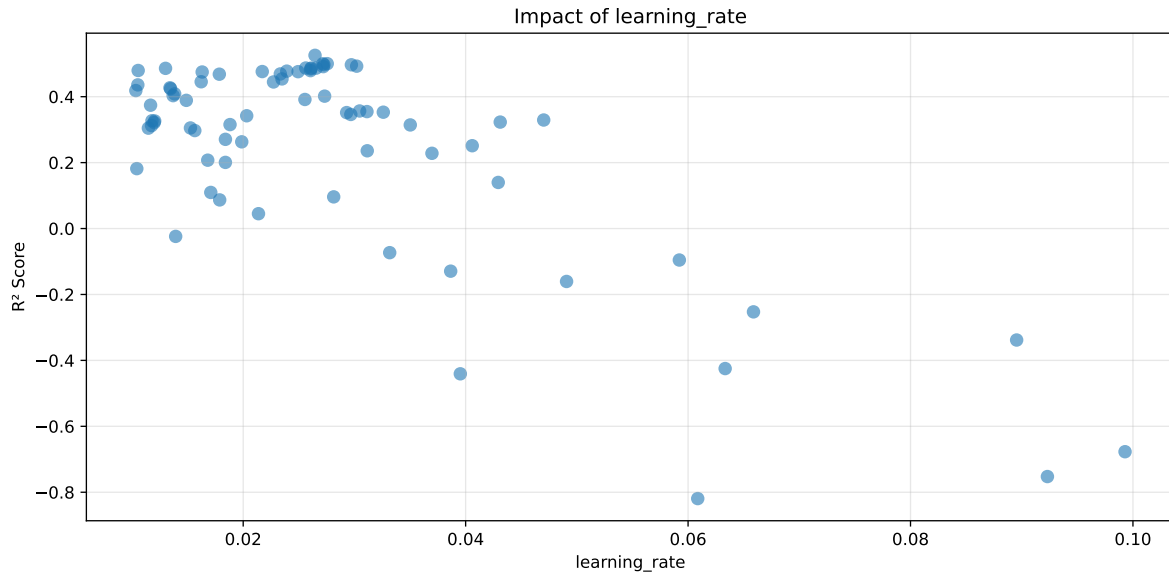
3.6.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.336, std=0.788, min=0.959, max=8.682
y_test: mean=2.345, std=0.834, min=1.062, max=13.845

```
===== TOP Importance FEATURES =====
feature importance r_umidity 46.779195 r_NH3 19.865994 r_CO2 7.848149 r_THI
6.904666 r_temperature 6.696466 r_NH3_fft_8h_fft_power1 4.637506 r_tempera-
ture_fft_8h_fft_power2 2.458428 r_umidity_fft_8h_fft_power2 1.925068 r_umid-
ity_fft_8h_fft_power1 1.054503 r_NH3_fft_8h_fft_f2_cph 0.884187 r_NH3_fft_8h_fft_power2
0.511616 r_temperature_fft_8h_fft_f2_cph 0.257131 r_umidity_fft_8h_fft_period3_h
0.093502 r_NH3_fft_8h_fft_period3_h 0.062641 r_umidity_fft_8h_fft_period1_h 0.020948
r_umidity_fft_8h_fft_period2_h 0.000000 r_NH3_fft_8h_fft_period1_h 0.000000
```

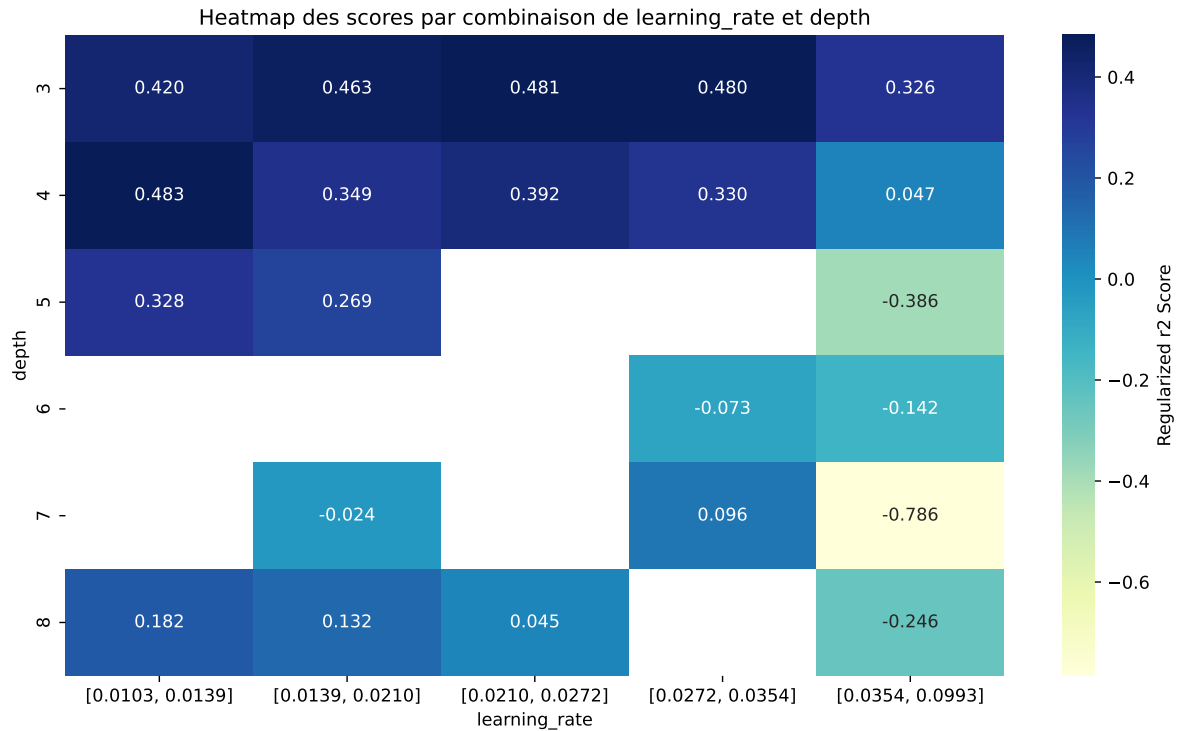


Paramètres avec >5% d'importance : ['learning_rate', 'depth']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5452	0.5255	0.45	0.0093	0.5255	0.4544	-0.071	0.8648	1.0376
2	0.5154	0.5002	0.4319	0.0107	0.5002	0.4413	-0.0589	0.8822	1.0303
3	0.5145	0.4999	0.428	0.0104	0.4999	0.4428	-0.0572	0.8857	1.0291
4	0.5123	0.4971	0.4266	0.0095	0.4971	0.4431	-0.054	0.8914	1.0307
5	0.5126	0.4968	0.4285	0.0112	0.4968	0.4362	-0.0606	0.8781	1.0318
6	0.5111	0.4926	0.4248	0.0108	0.4926	0.4362	-0.0564	0.8855	1.0376
7	0.5065	0.4913	0.4221	0.0095	0.4913	0.429	-0.0623	0.8732	1.031
8	0.502	0.487	0.4197	0.0101	0.487	0.4299	-0.0571	0.8828	1.0309
9	0.5	0.4868	0.42	0.0093	0.4868	0.426	-0.0607	0.8752	1.0272
10	0.5019	0.4867	0.4199	0.011	0.4867	0.4222	-0.0645	0.8676	1.0313
---	---	---	---	---	---	---	---	---	---
75	0.9785	0.6789	0.5955	0.0165	-0.8192	0.6184	-0.0604	0.911	1.4414

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.026457333010513607, 'reg__l2_leaf_reg': 2.2818458908723773, 'reg__bagging_temperature': 4.920127996223547, 'reg__random_strength': 0.170934703518272, 'reg__subsample': 0.639265564153767, 'reg__min_data_in_leaf': 46, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8982675089455401, 'reg__border_count': 128, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 360.4813218435702}

Rank 2: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02755712121293368, 'reg__l2_leaf_reg': 13.915653683596393, 'reg__bagging_temperature': 4.106873720700155, 'reg__random_strength': 3.1648140180068944, 'reg__subsample': 0.5811903693250449, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.7654138234393032, 'reg__border_count': 93, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3329.056241256755}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.027162555400773416, 'reg__l2_leaf_reg': 1.3376894926062883, 'reg__bagging_temperature': 4.102861887814203, 'reg__random_strength': 3.1826626383316228, 'reg__subsample': 0.5750508066441586, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6848980295907307, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4561.263703262961}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.027259772906624756, 'reg__l2_leaf_reg': 14.204694111765633, 'reg__bagging_temperature': 4.064849315930877, 'reg__random_strength': 3.1426634938241267, 'reg__subsample': 0.5781264899142474, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.699068071091417, 'reg__border_count': 106, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3289.220286526348}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.029726674798557406, 'reg__l2_leaf_reg': 8.975215266571409, 'reg__bagging_temperature': 2.875669797842496, 'reg__random_strength': 4.488650856412639, 'reg__subsample': 0.5428223723862889, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8612345028159525, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4721.31061865469}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.03020275264703438, 'reg__l2_leaf_reg': 7.847061404185581, 'reg__bagging_temperature': 2.9336550122355947, 'reg__random_strength': 4.5177893038423695, 'reg__subsample': 0.5448634064551372, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7618562926489135, 'reg__border_count': 85, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4840.658972642652}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02720026452369216, 'reg__l2_leaf_reg': 1.1965168755460294, 'reg__bagging_temperature': 2.8324686270285646, 'reg__random_strength': 4.991335464187569, 'reg__subsample': 0.6071736186976894,

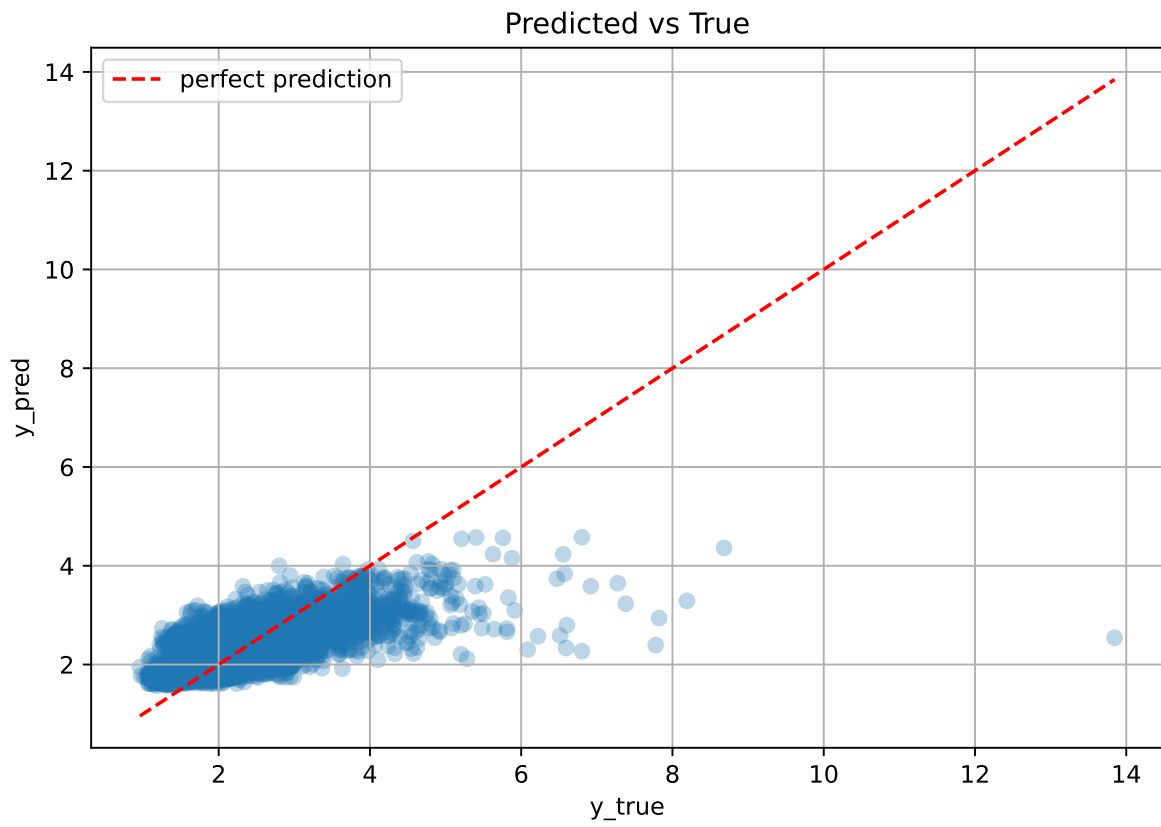
'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.880127574709027, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4581.19500298629}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.025626042906625564, 'reg__l2_leaf_reg': 3.3063901335325694, 'reg__bagging_temperature': 1.7605613951753472, 'reg__random_strength': 3.828768638175035, 'reg__subsample': 0.6098822893436106, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8815774469260806, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3218.036133548018}

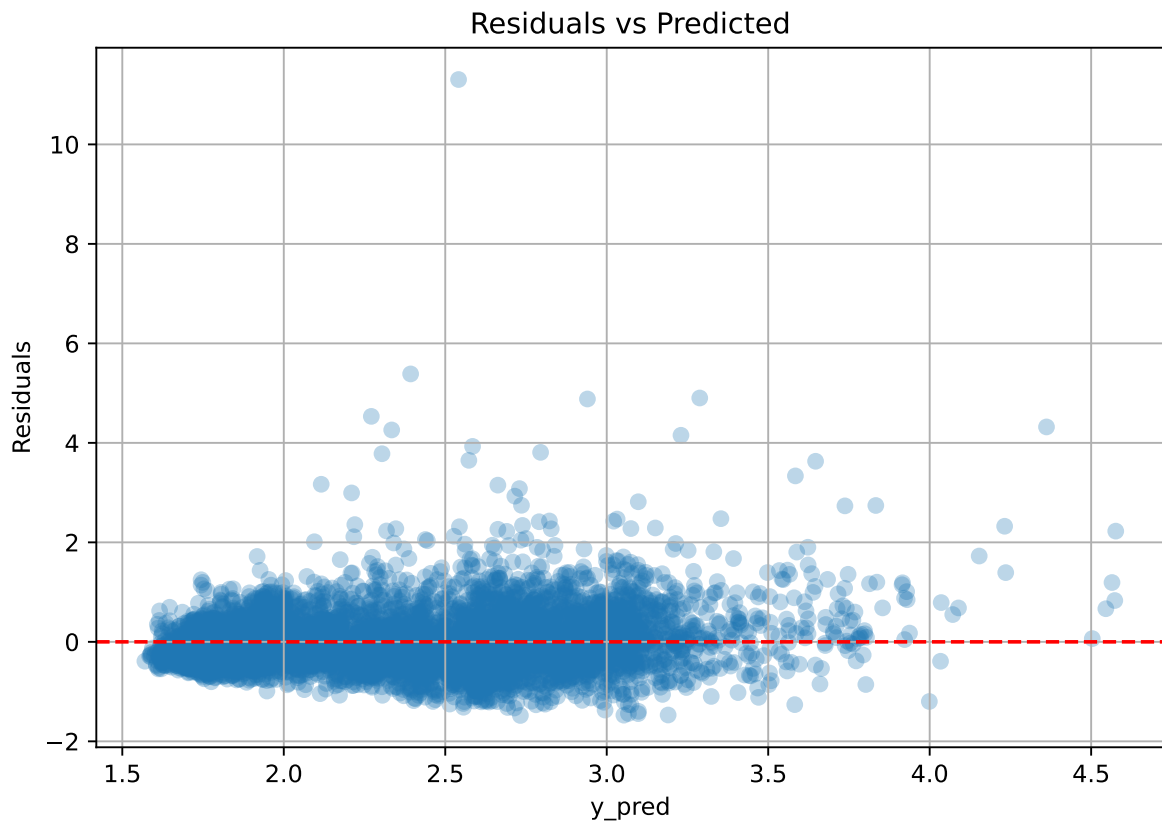
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.026112903788250097, 'reg__l2_leaf_reg': 11.383939302934575, 'reg__bagging_temperature': 3.5494450017408585, 'reg__random_strength': 3.814664817823076, 'reg__subsample': 0.6130125815557916, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7617947140101105, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4551.590577003819}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02656174743492917, 'reg__l2_leaf_reg': 1.1738585634056946, 'reg__bagging_temperature': 2.803261826722999, 'reg__random_strength': 4.98948402726059, 'reg__subsample': 0.6148086870150269, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8831563076129888, 'reg__border_count': 120, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3217.8878491813553}

3.6.2 Scatter



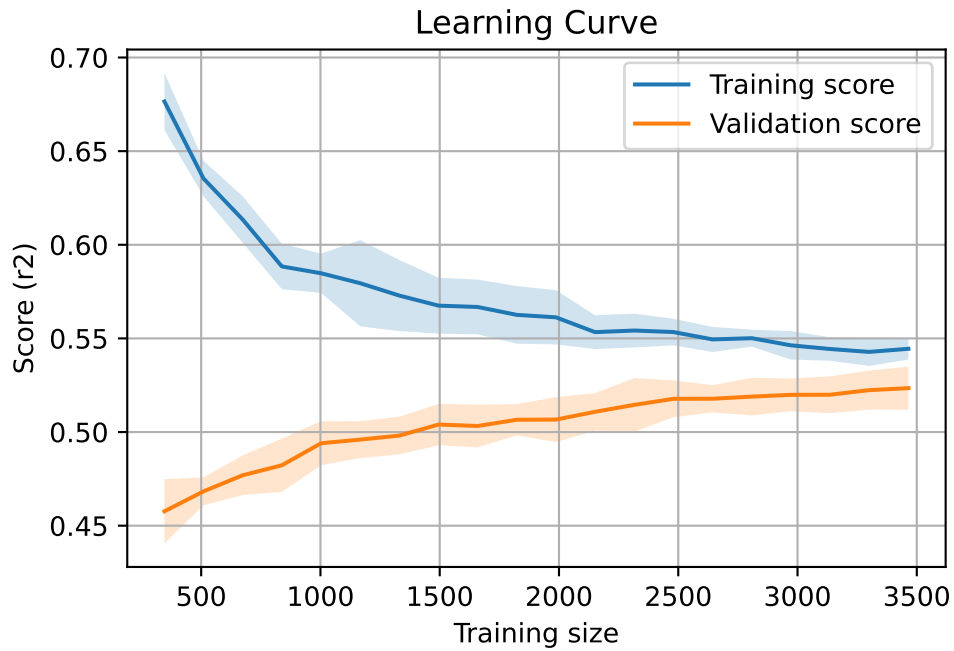
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7072, `Len(training)` : 4951,

`Len(training_real)` : 3466, `Len(test_of_training)` : 1485, `Len(test)` : 2121



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.393	0.386	0.009	0.336	0.338	-0.048	0.876	1.02	nan
LGBM	0.538	0.513	0.009	0.444	0.468	-0.045	0.912	1.048	0.5134
CatBoost	0.545	0.526	0.009	0.45	0.454	-0.071	0.865	1.038	0.5255

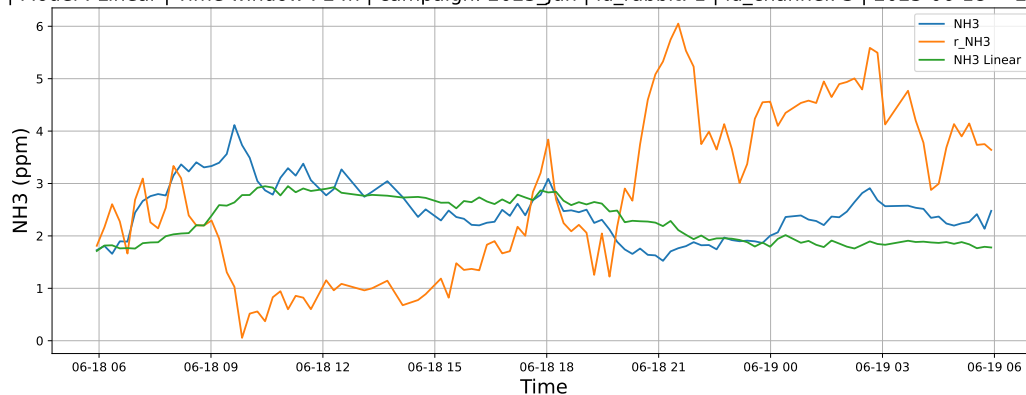
4 Plot

4.1 Standalone plots

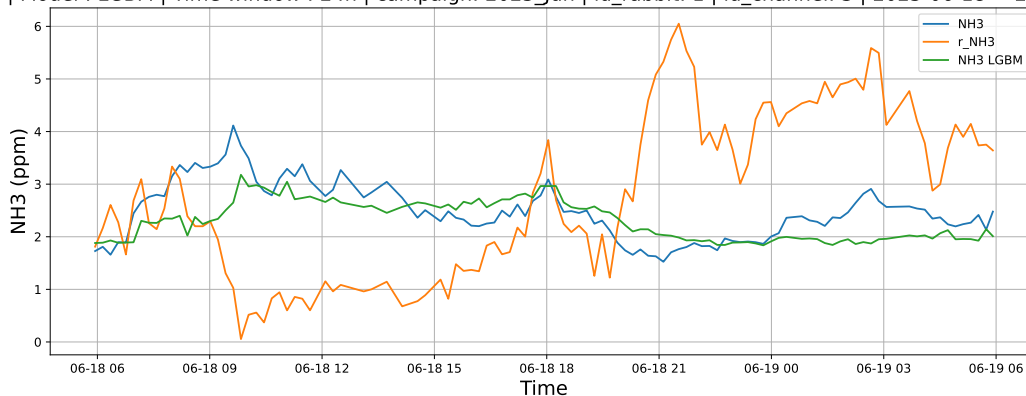
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

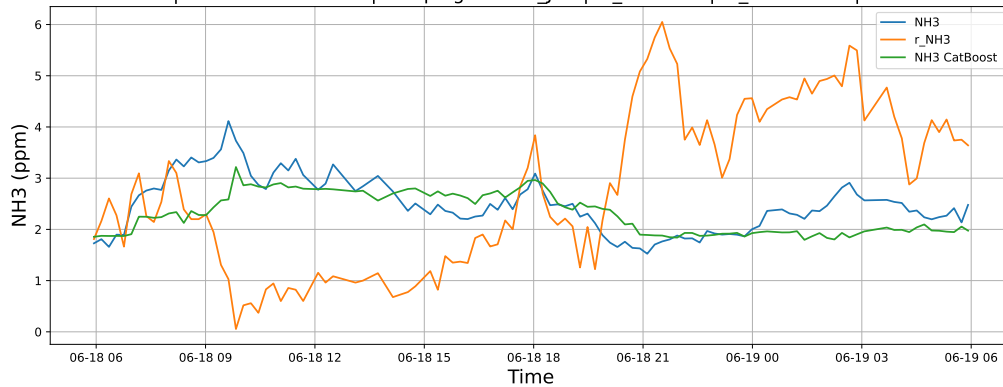
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

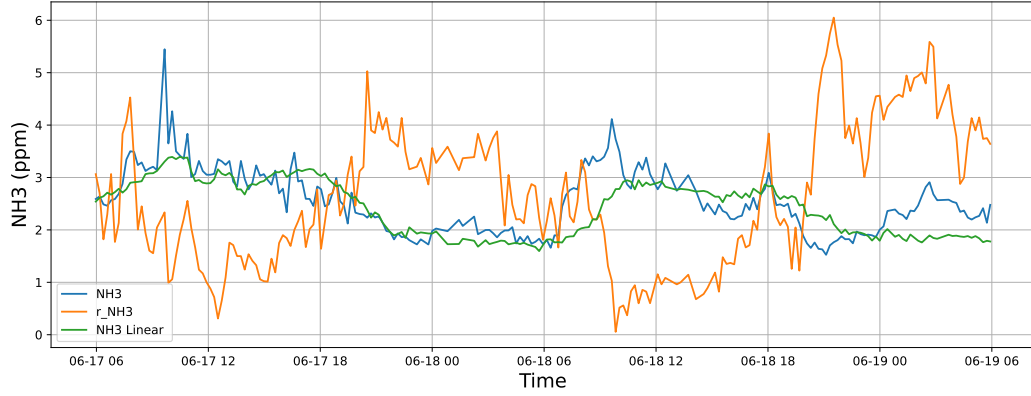


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

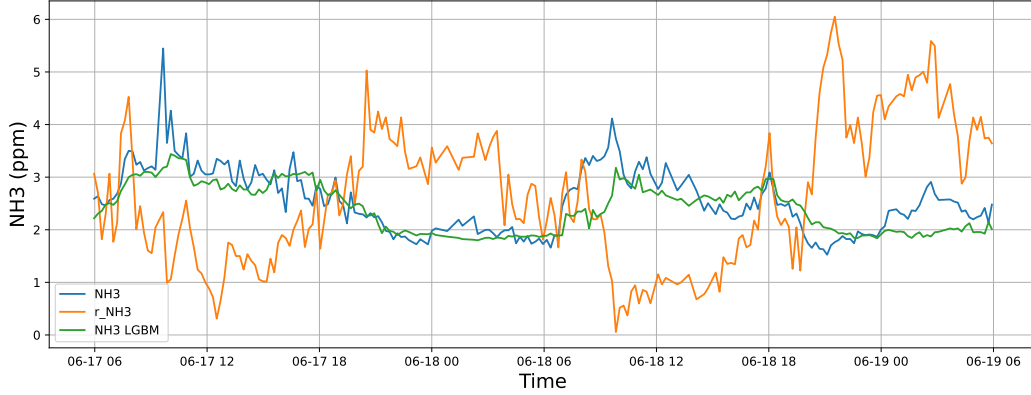


4.1.1.2 Time window : 48

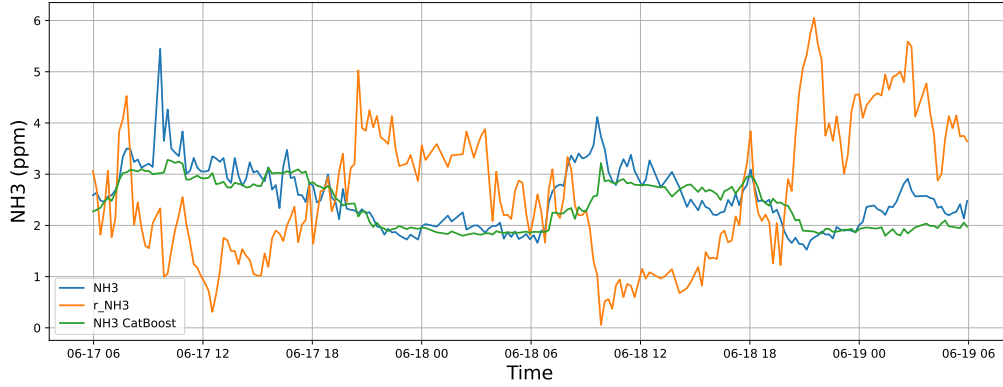
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

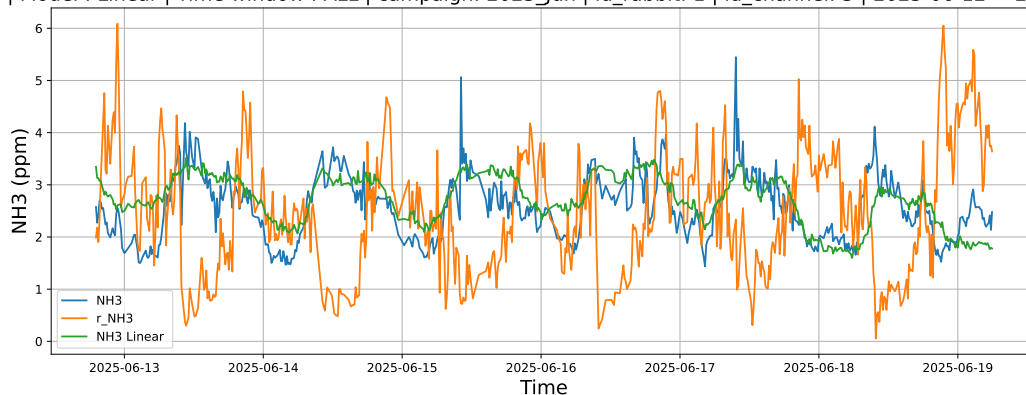


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

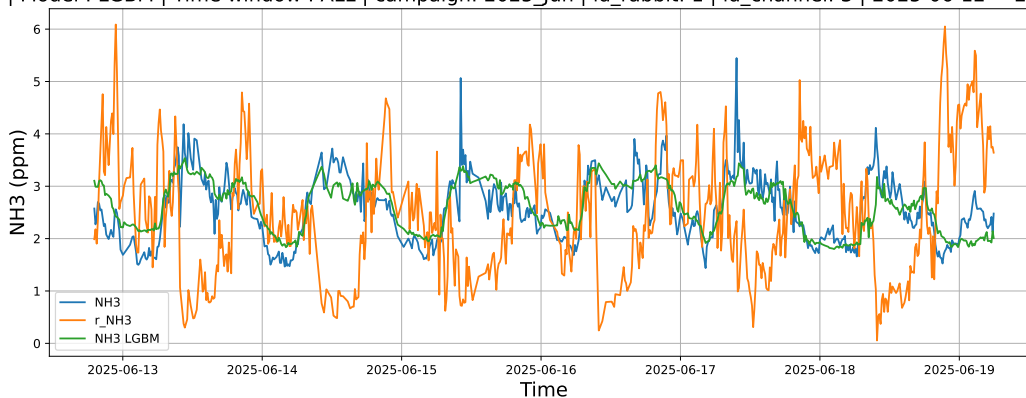


4.1.1.3 Time window : ALL

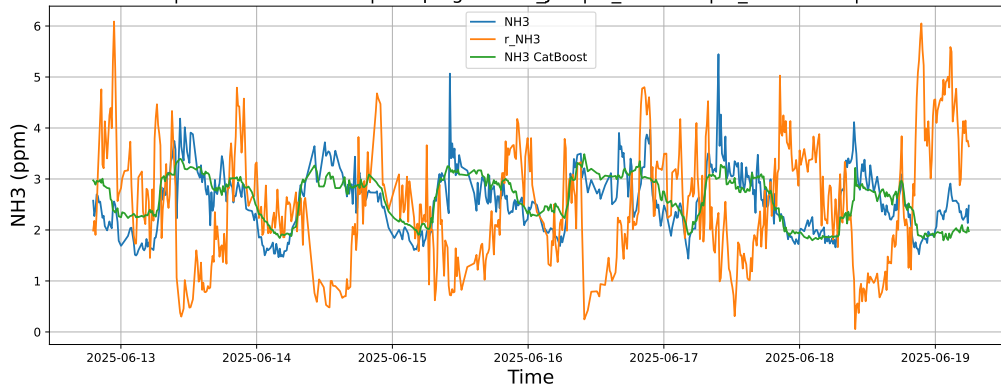
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



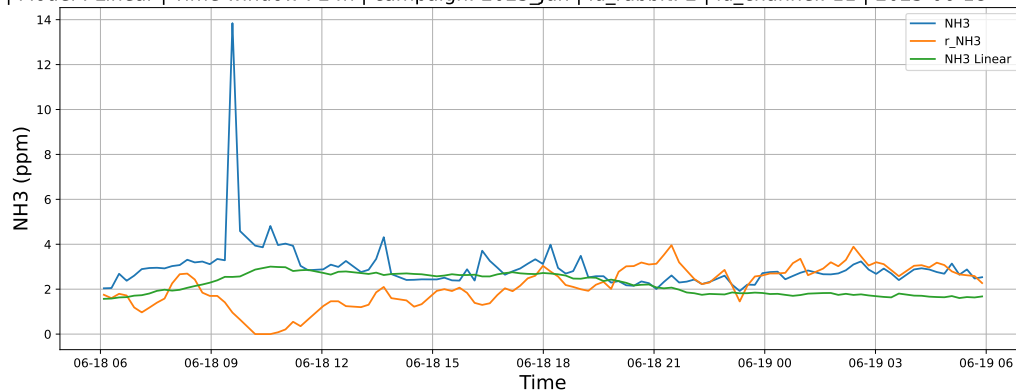
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



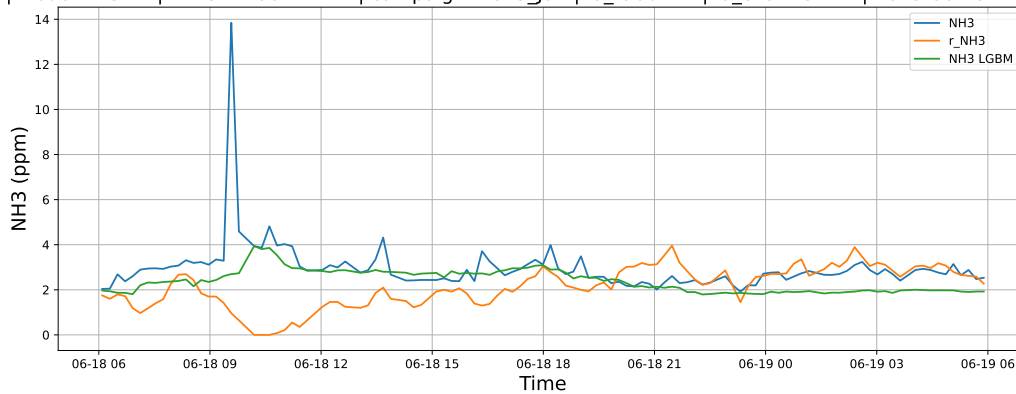
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

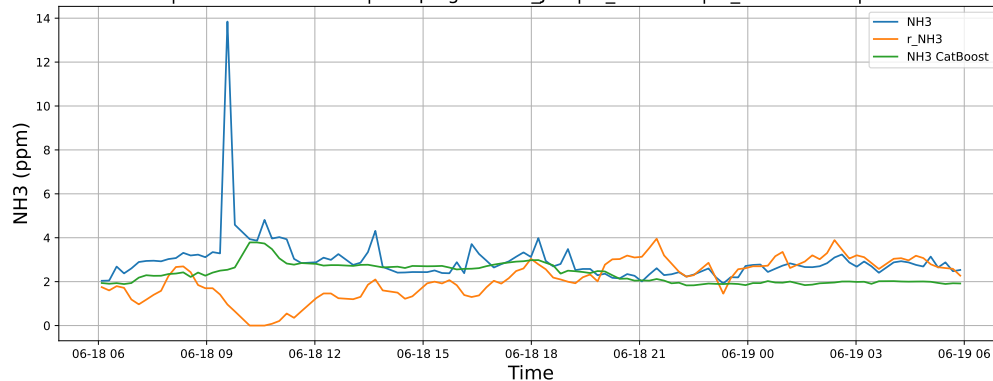
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

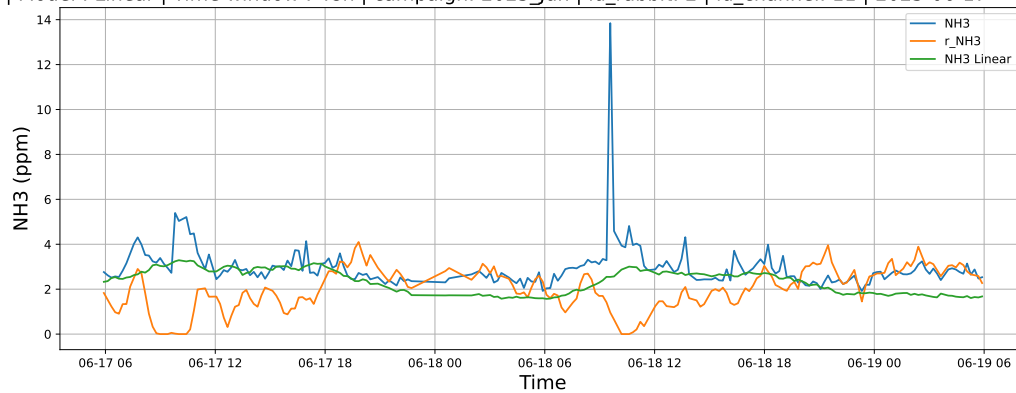


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

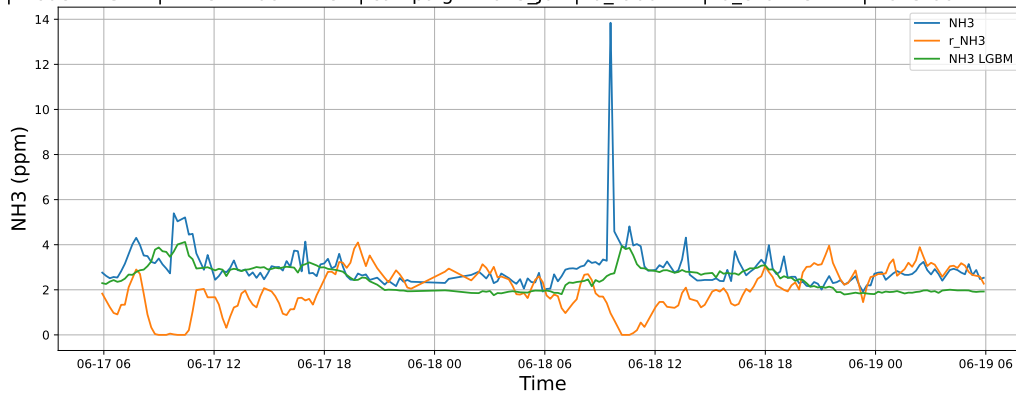


4.1.2.2 Time window : 48

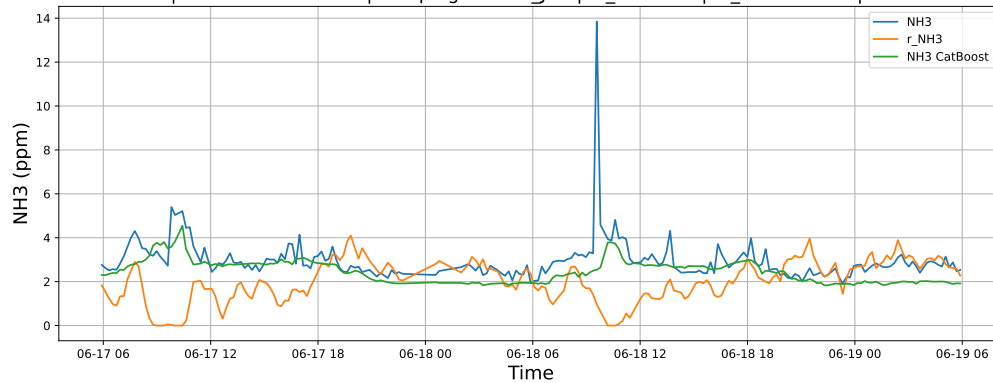
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

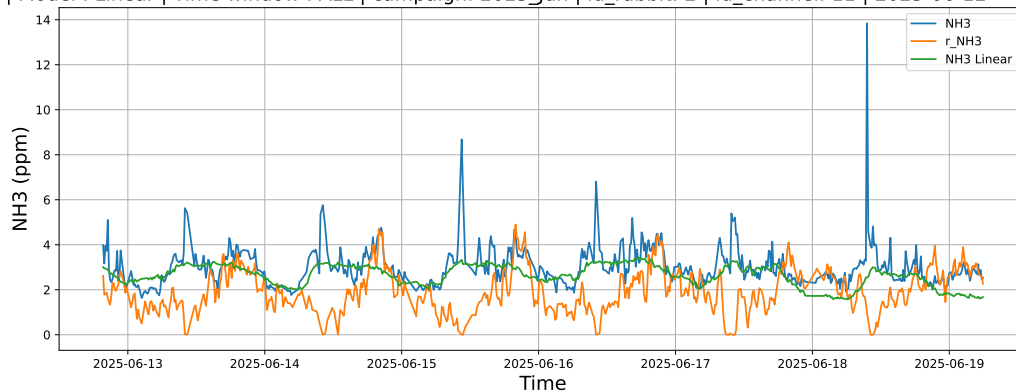


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

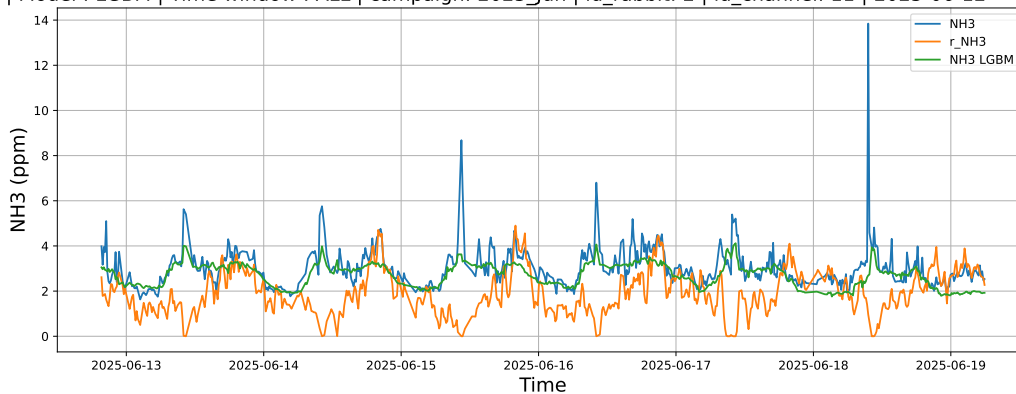


4.1.2.3 Time window : ALL

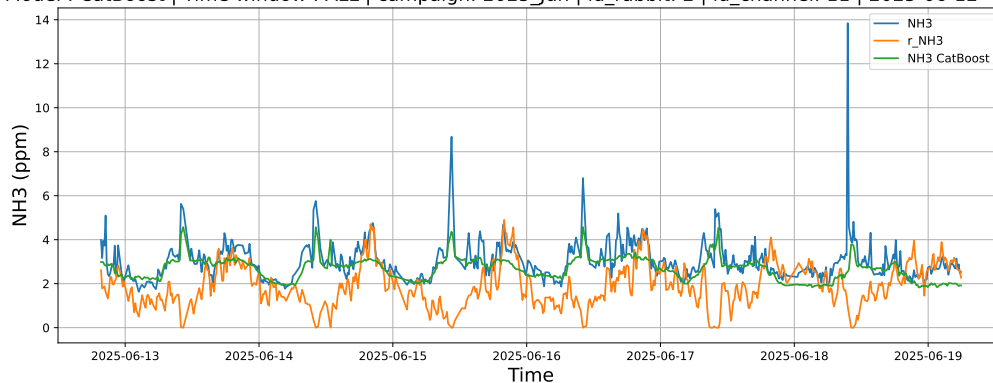
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



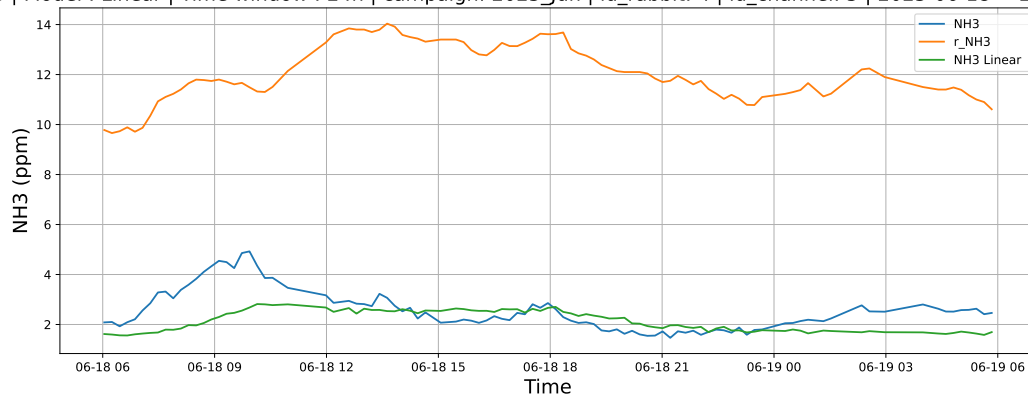
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



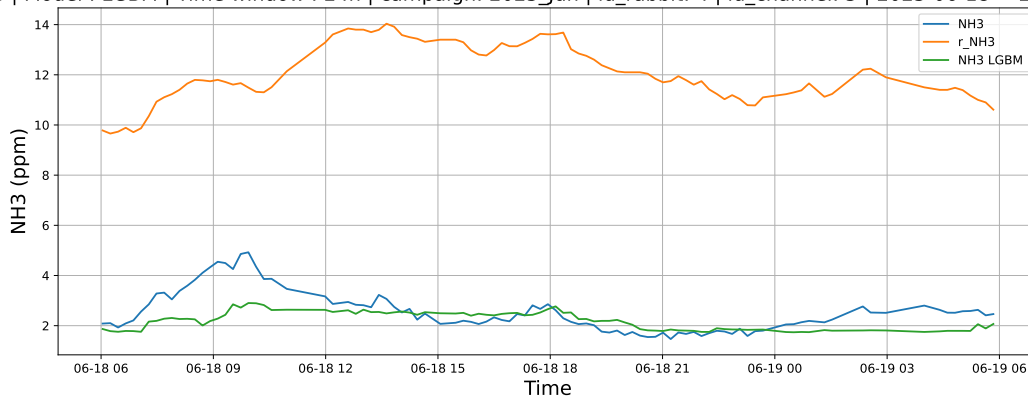
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

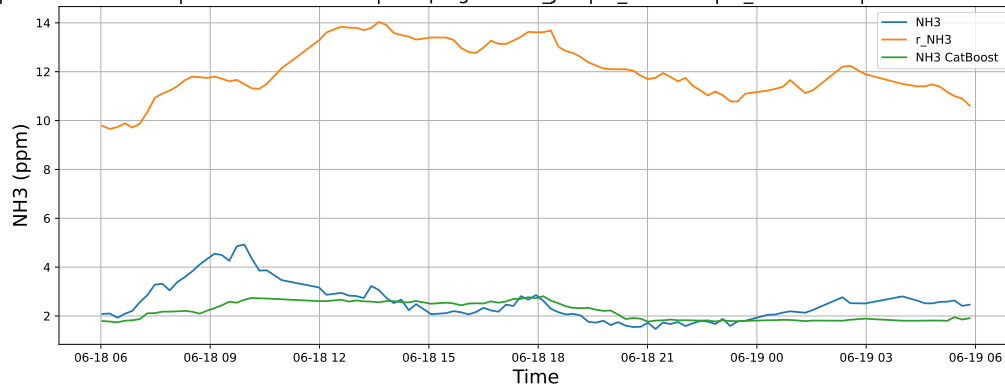
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

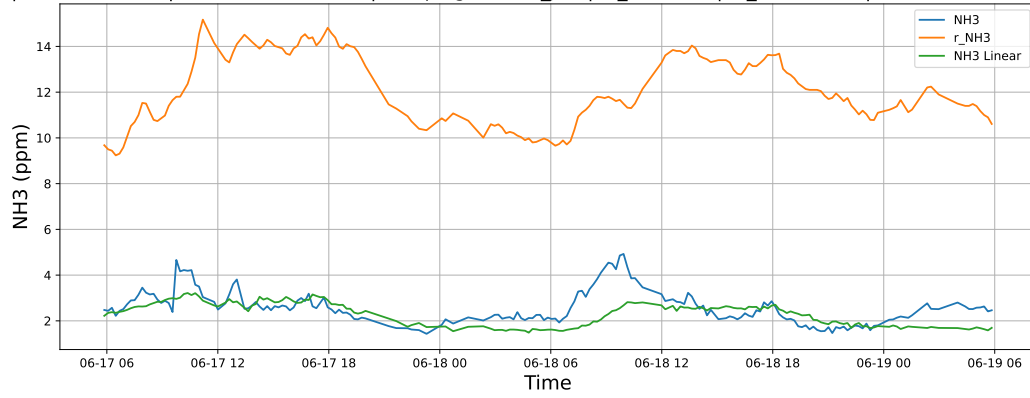


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

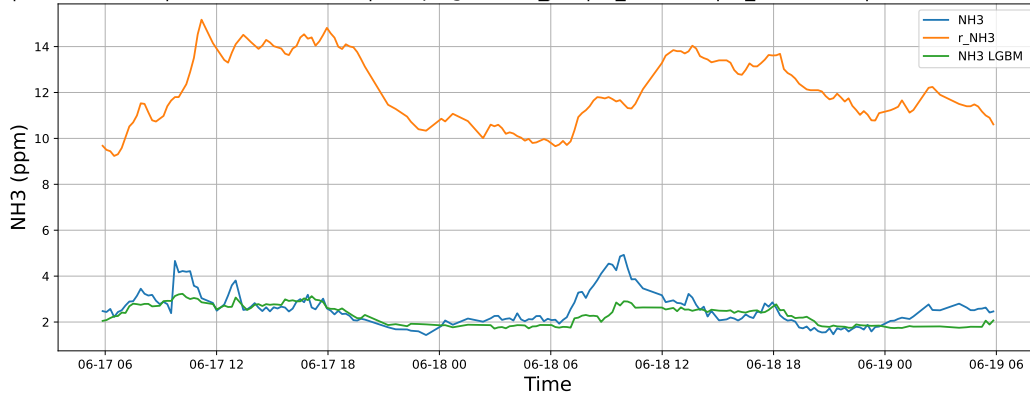


4.1.3.2 Time window : 48

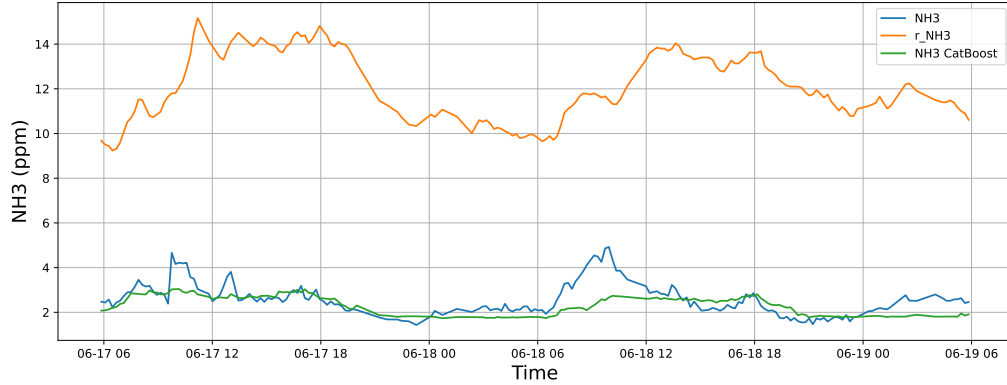
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

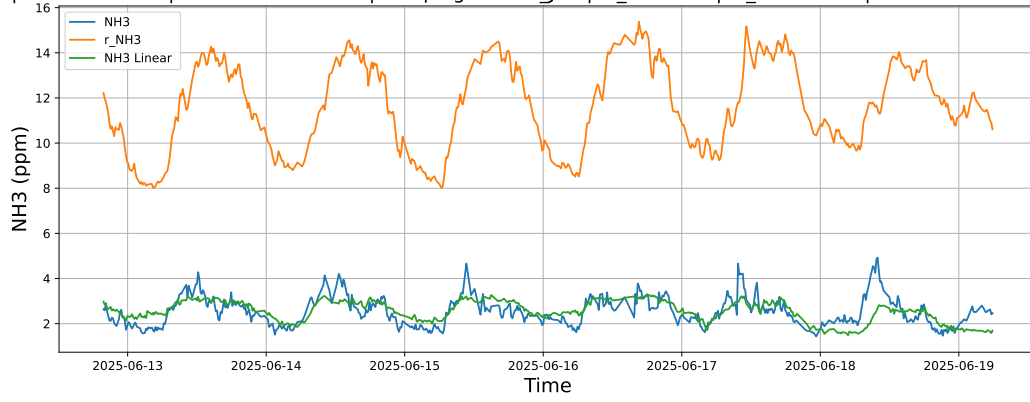


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

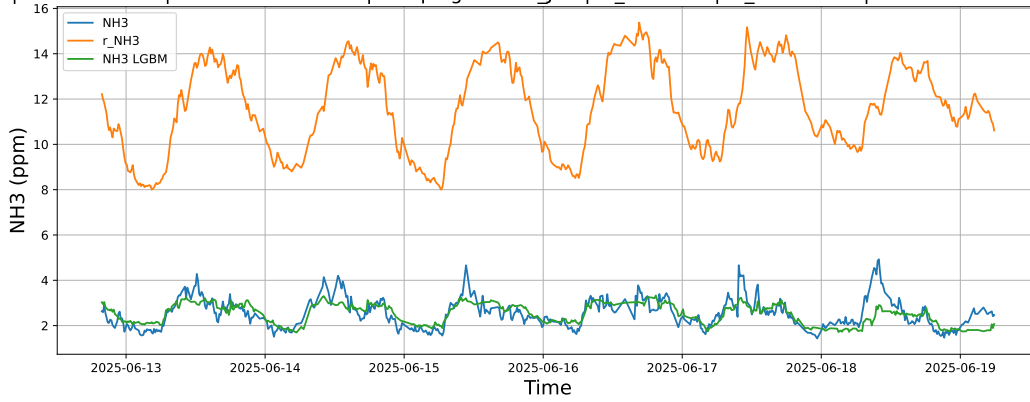


4.1.3.3 Time window : ALL

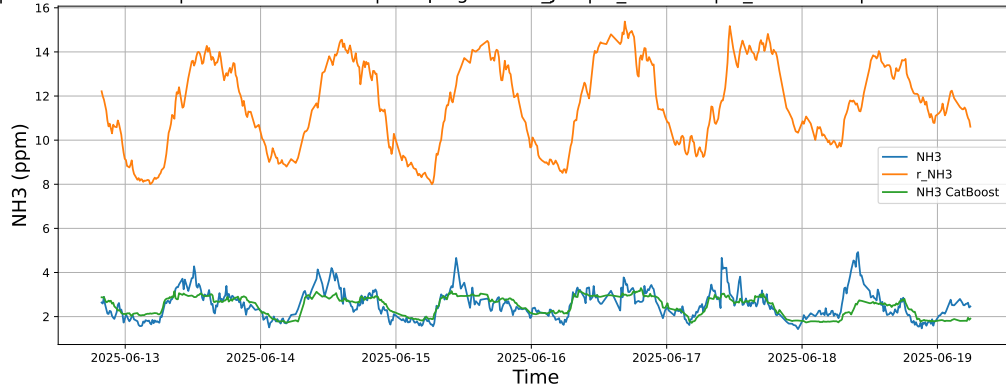
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



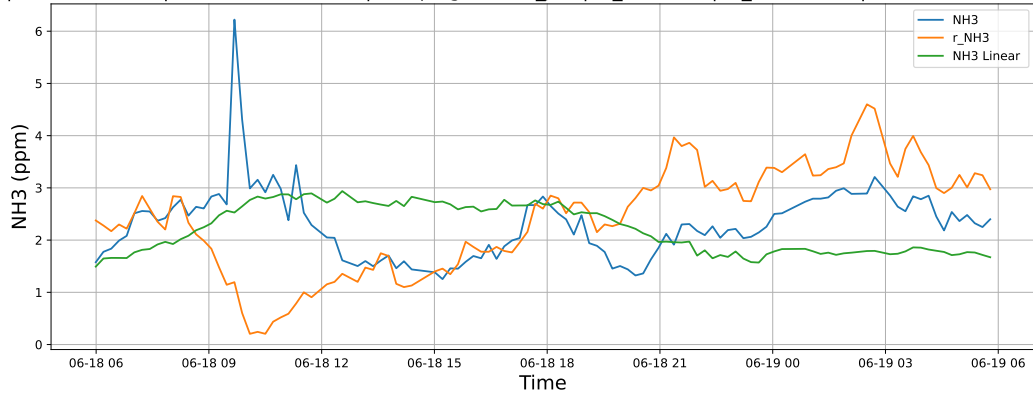
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



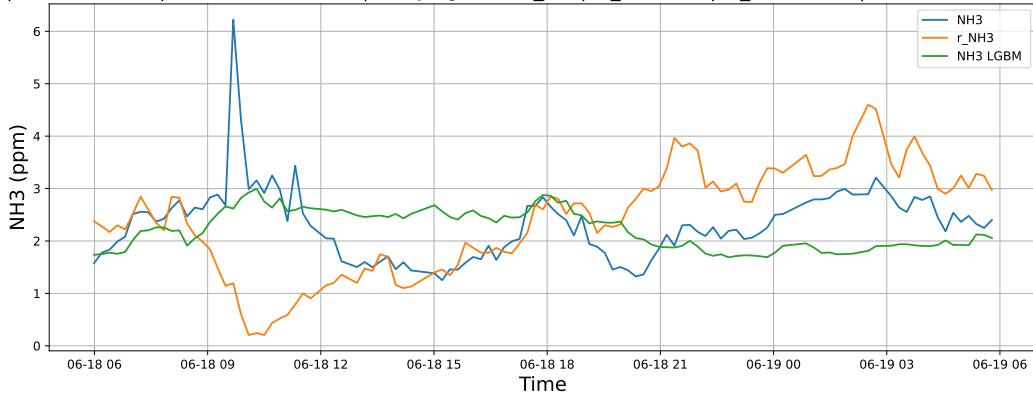
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

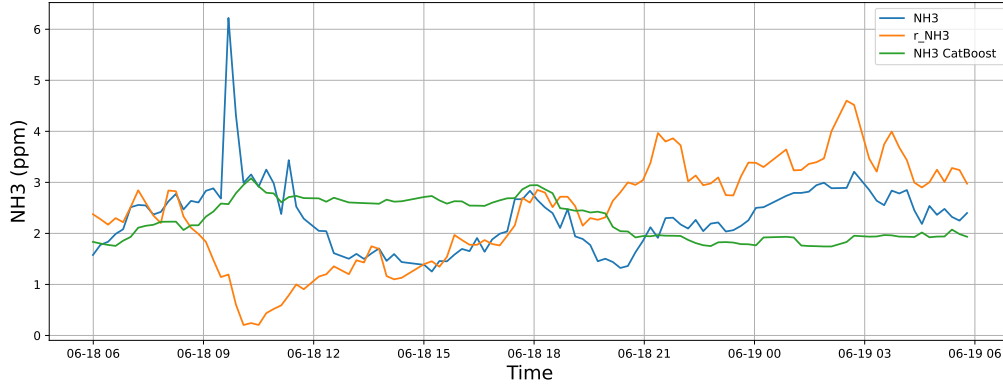
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

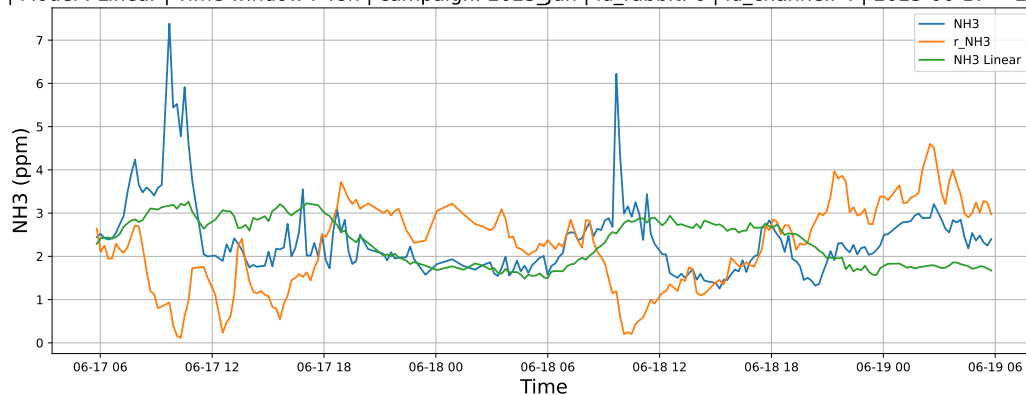


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

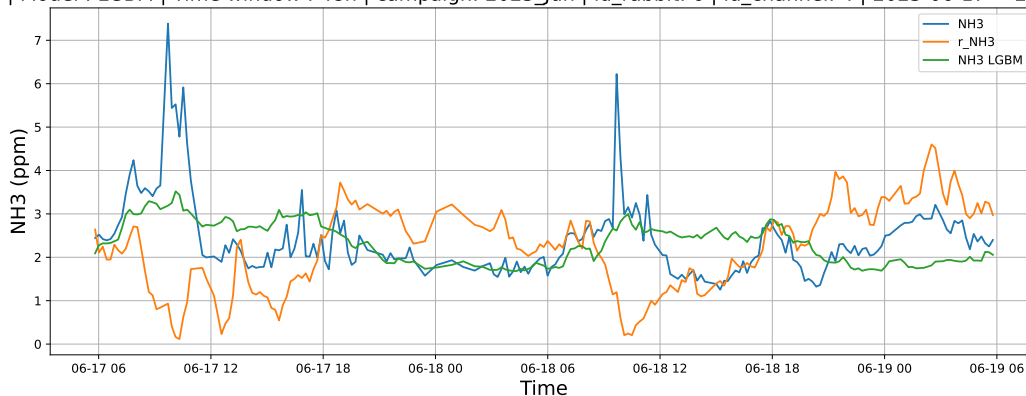


4.1.4.2 Time window : 48

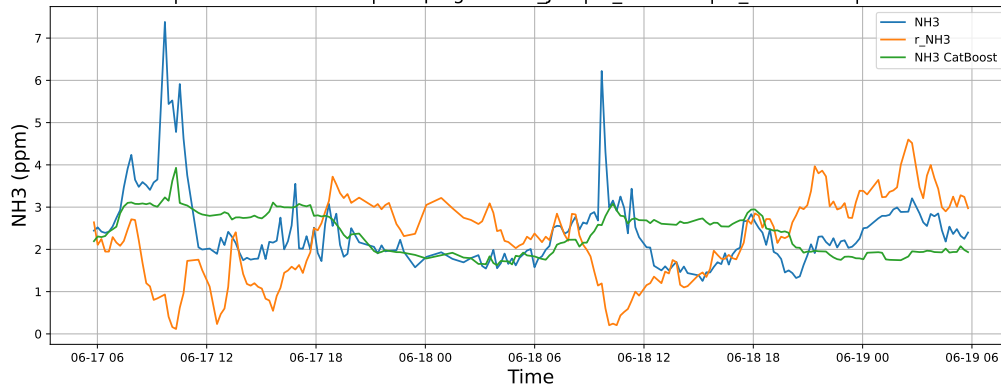
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

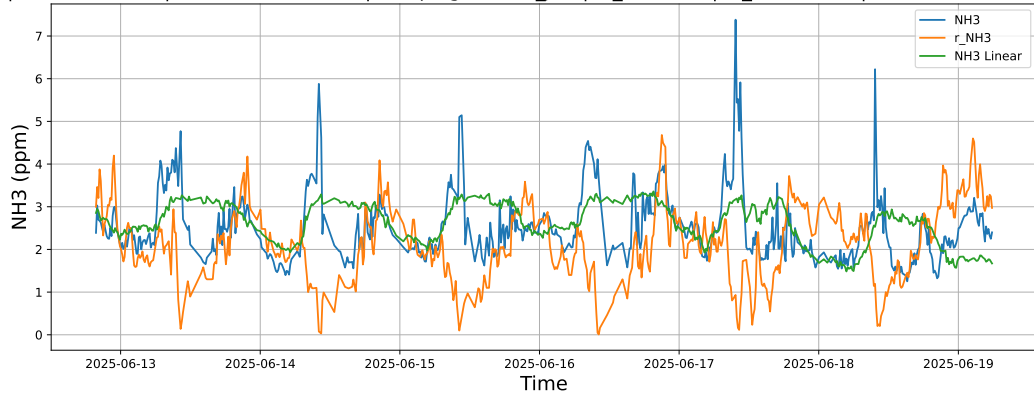


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

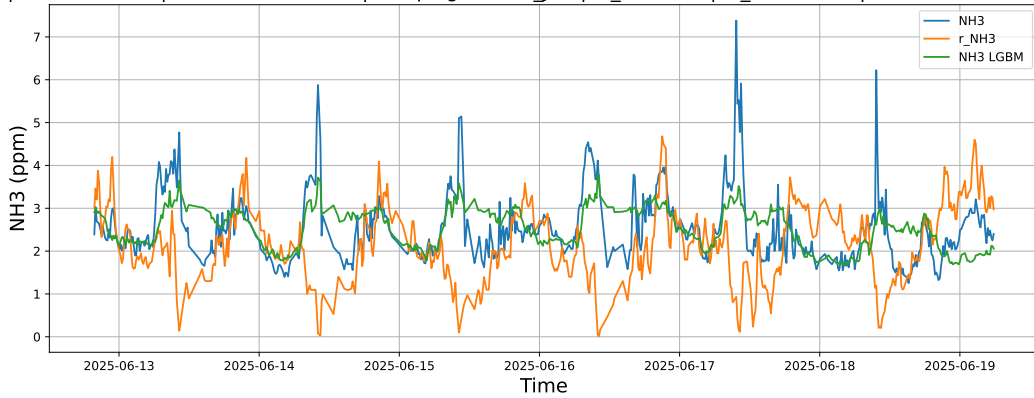


4.1.4.3 Time window : ALL

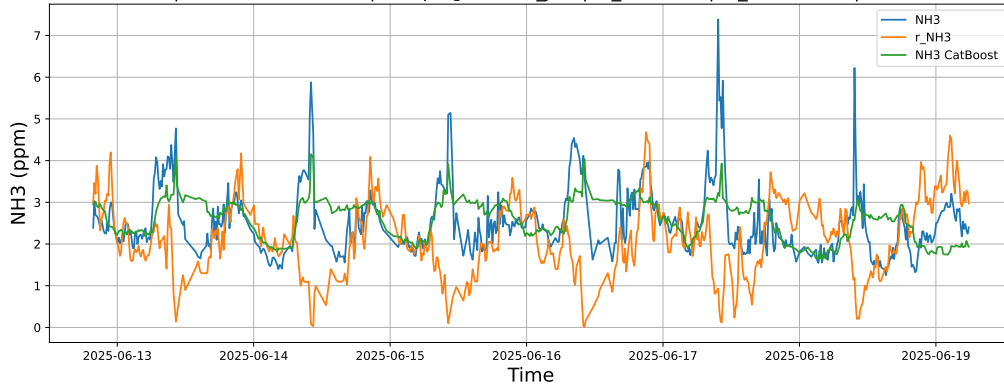
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



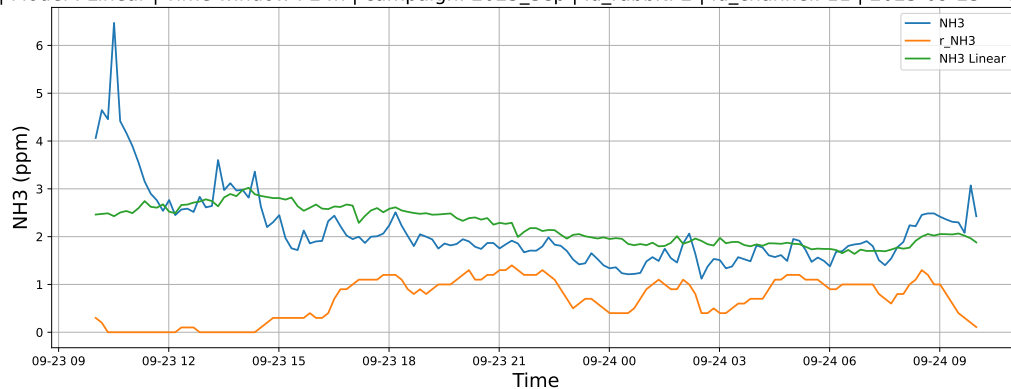
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



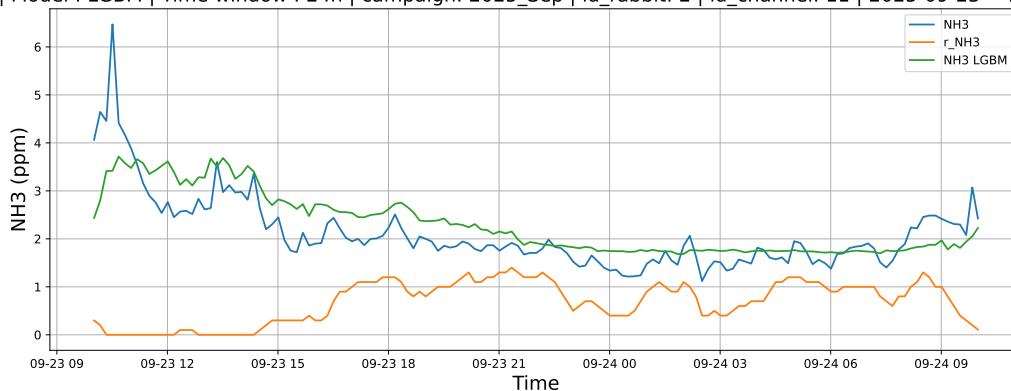
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

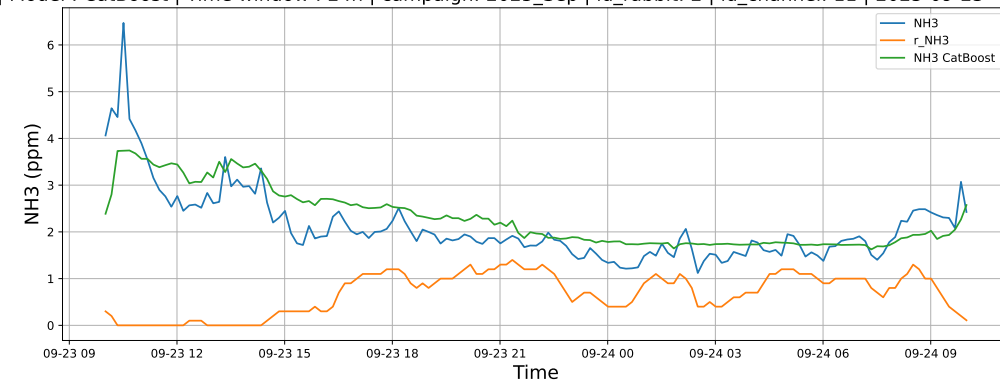
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

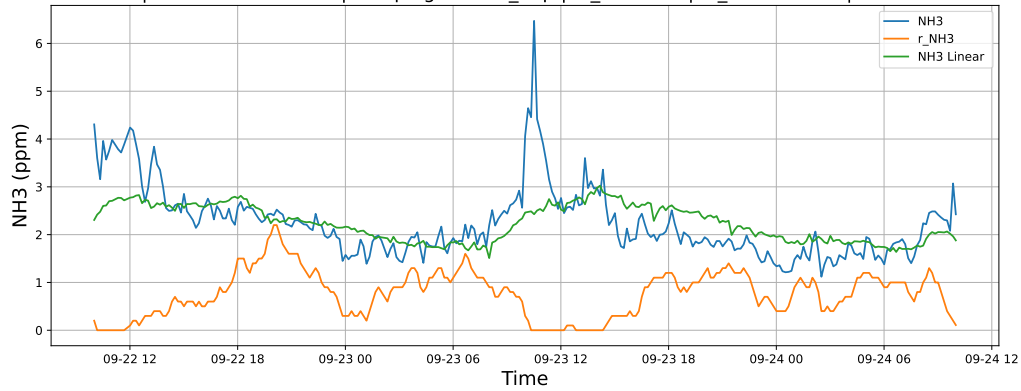


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

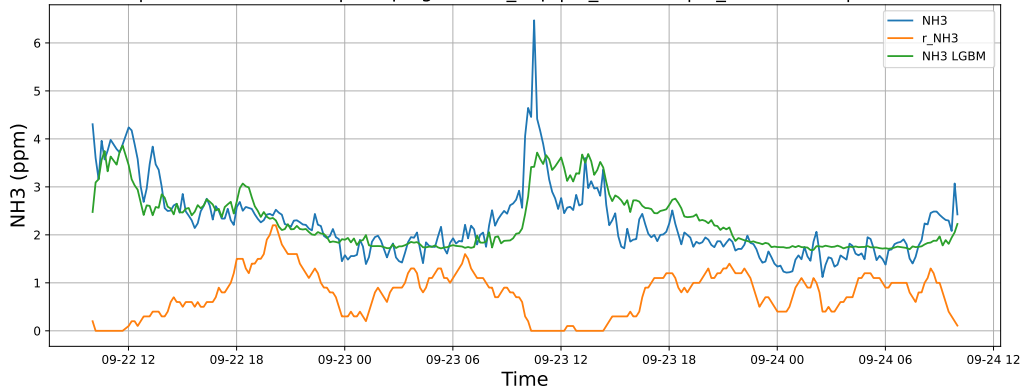


4.1.5.2 Time window : 48

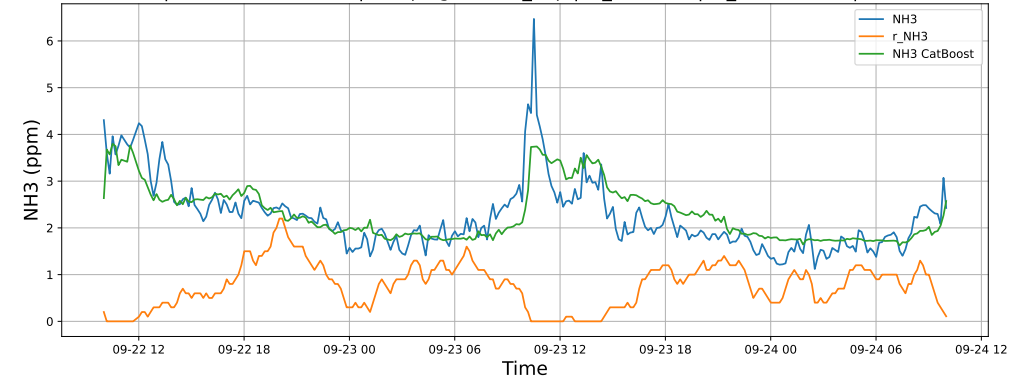
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

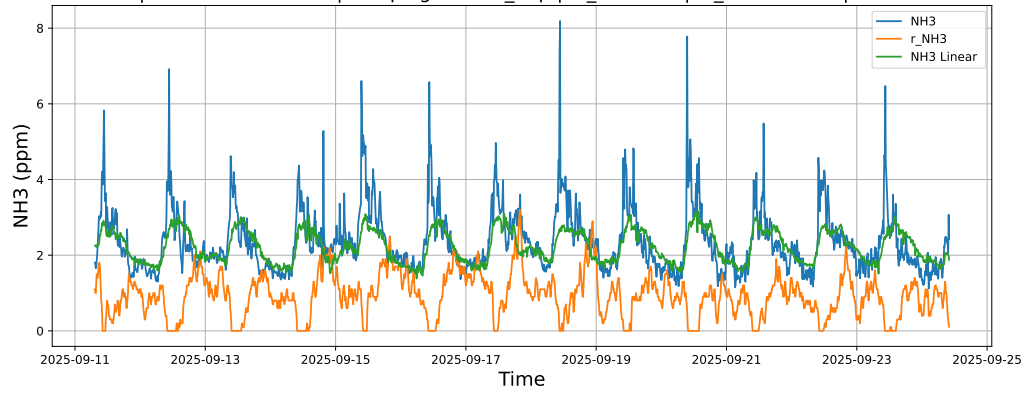


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

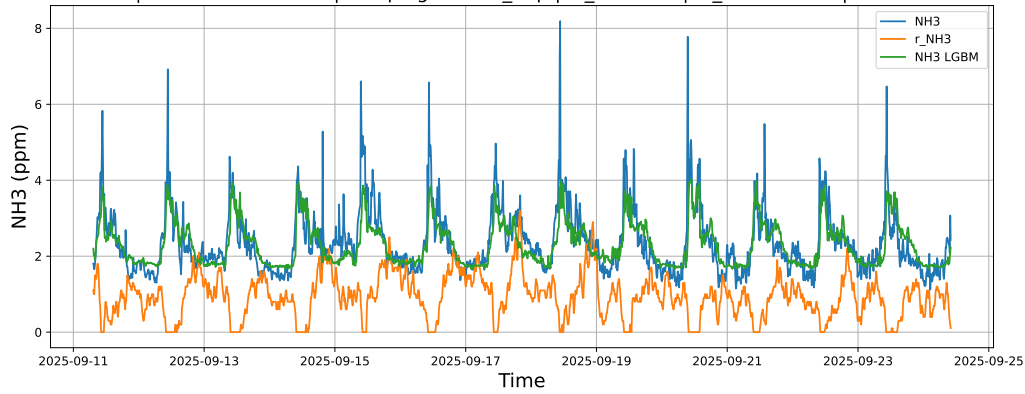


4.1.5.3 Time window : ALL

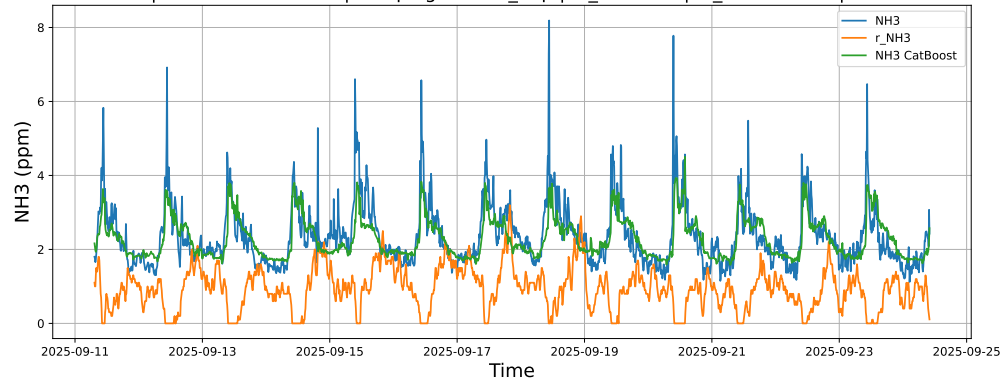
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



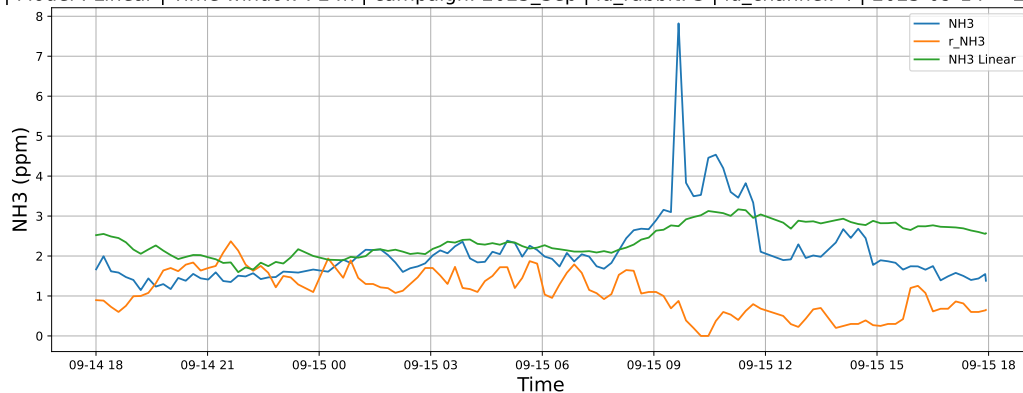
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



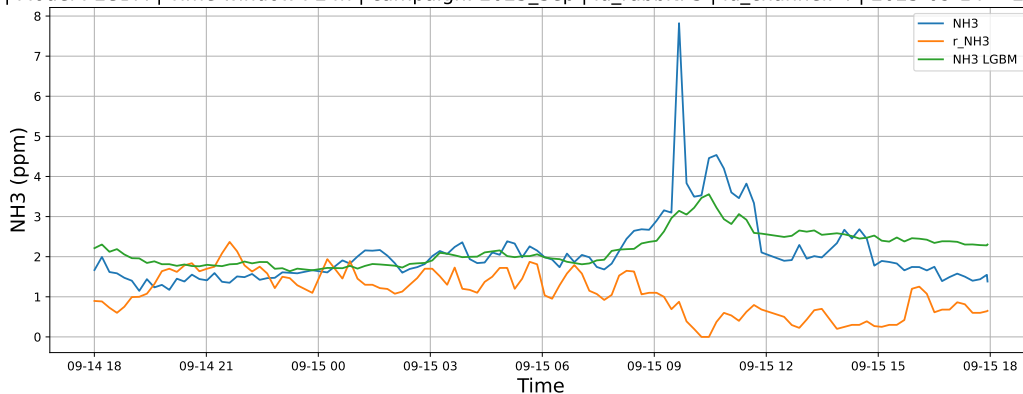
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

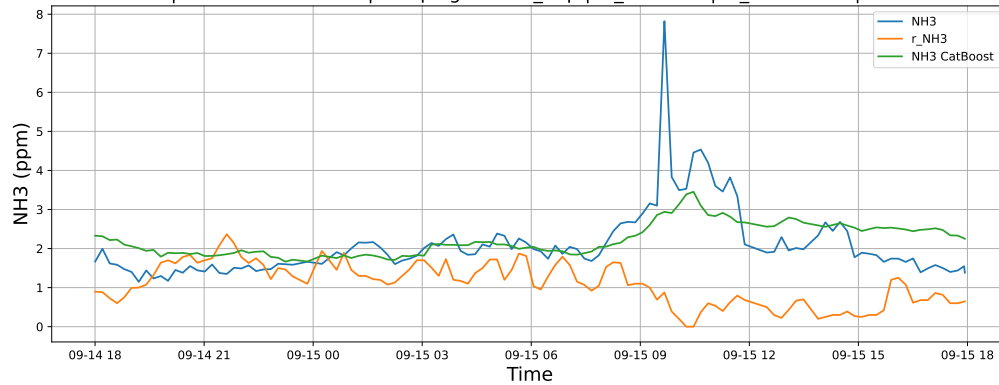
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

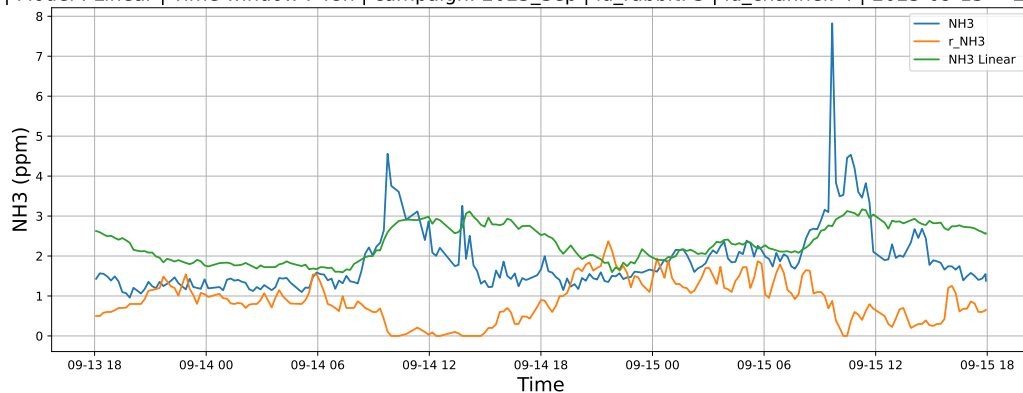


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

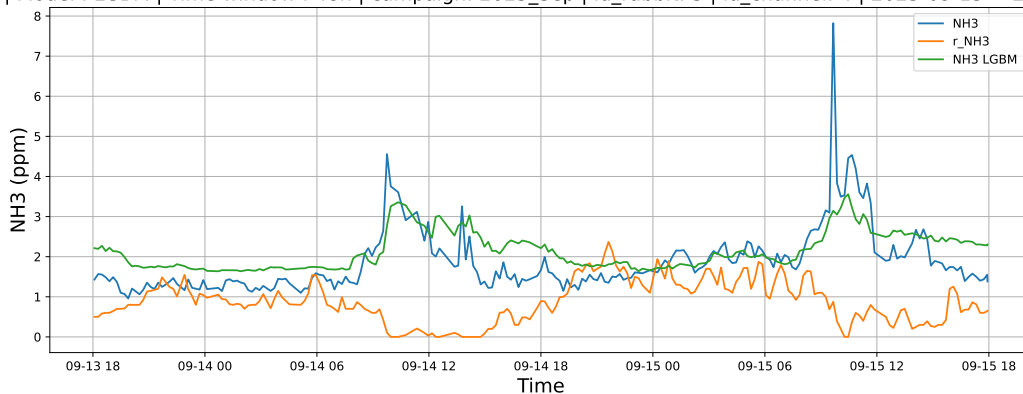


4.1.6.2 Time window : 48

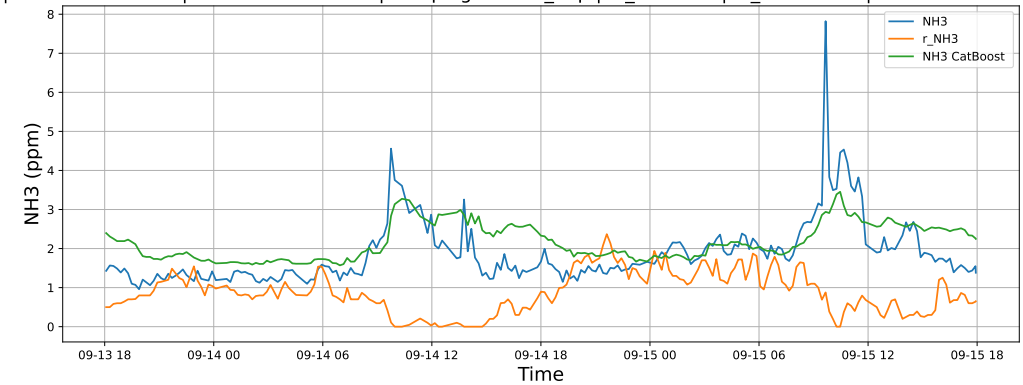
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

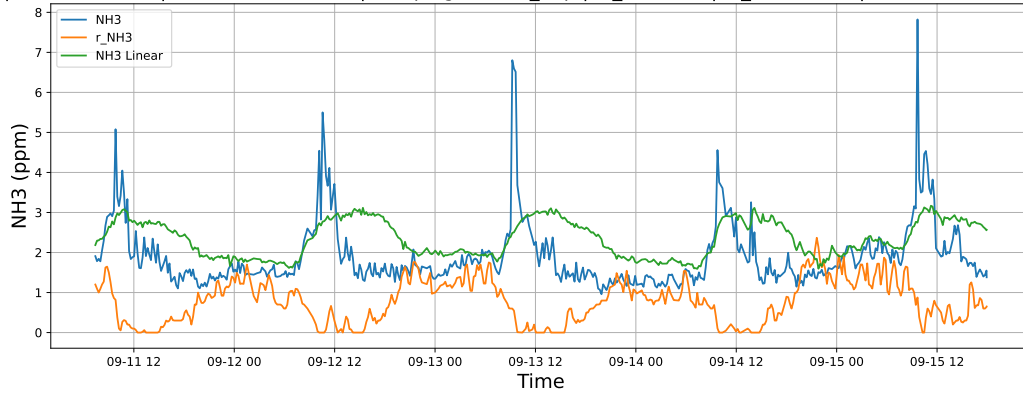


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

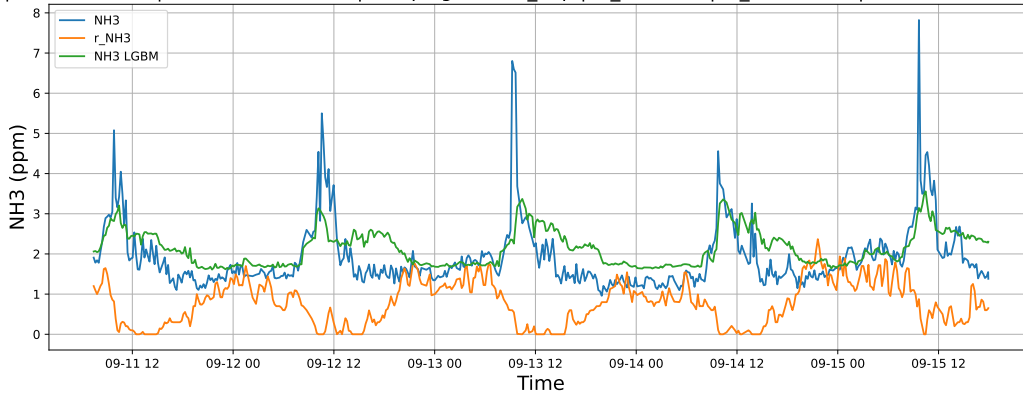


4.1.6.3 Time window : ALL

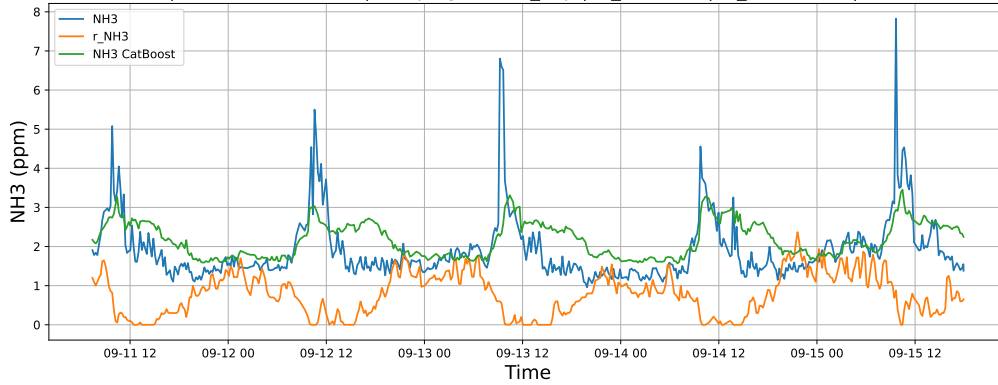
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



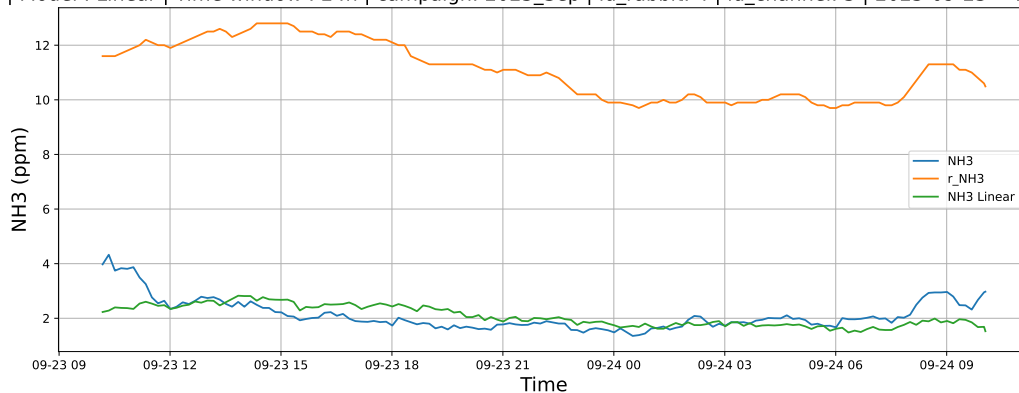
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



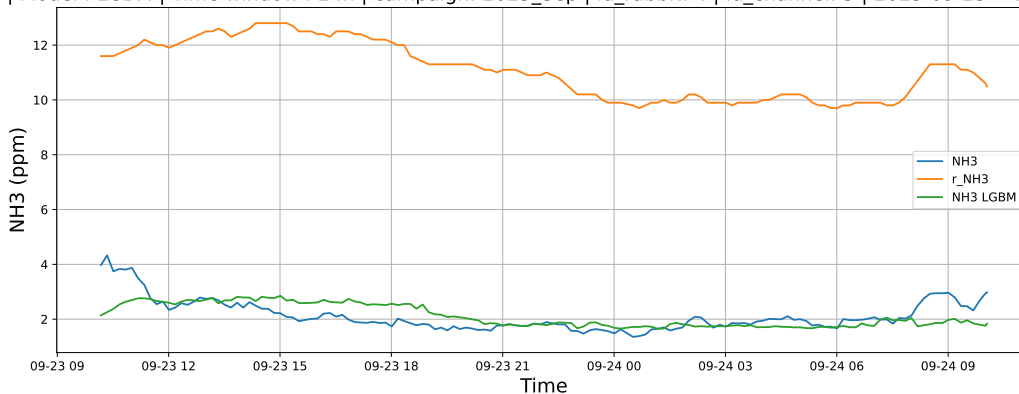
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

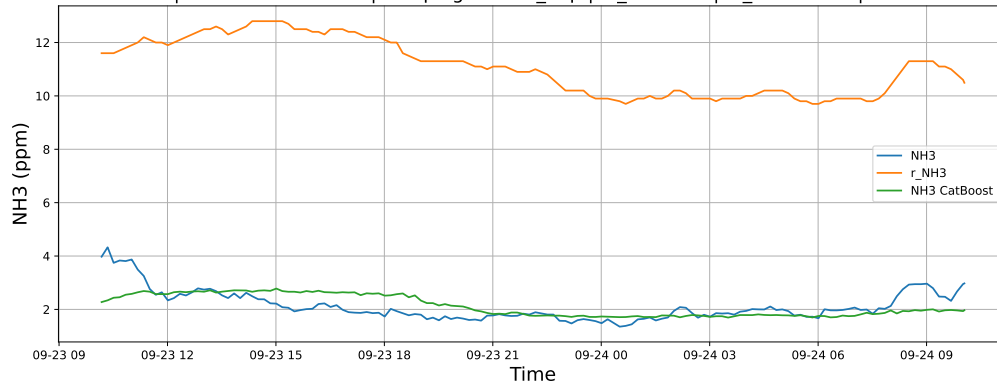
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

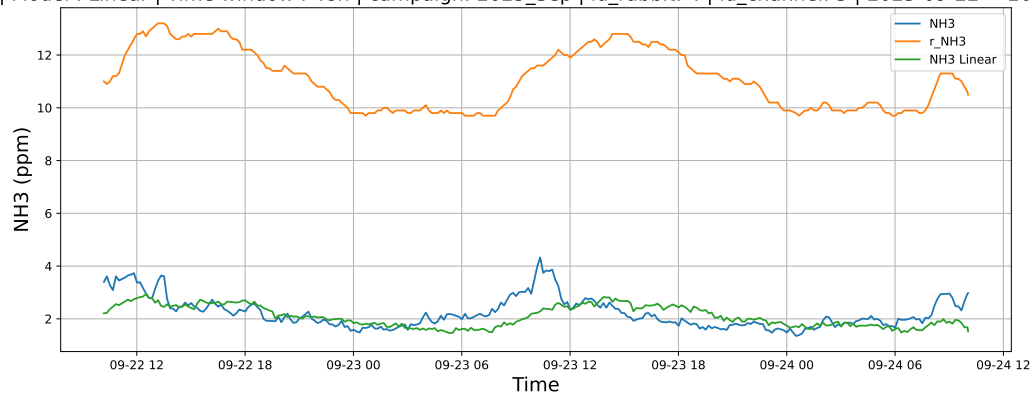


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

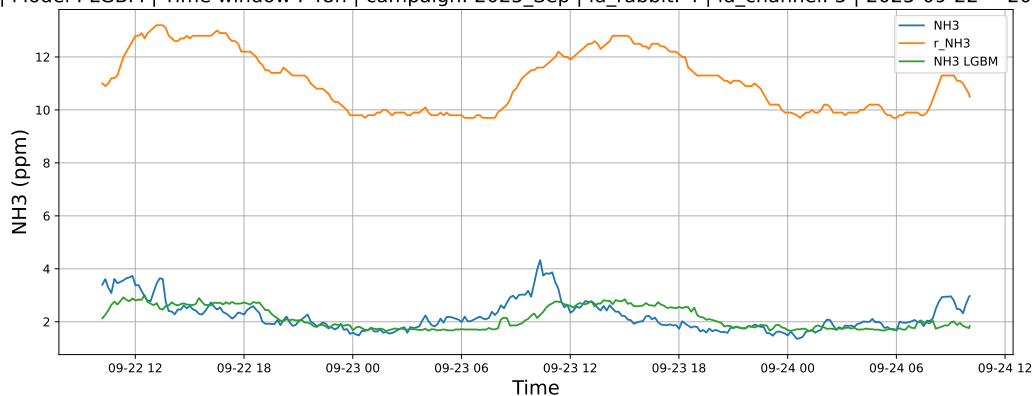


4.1.7.2 Time window : 48

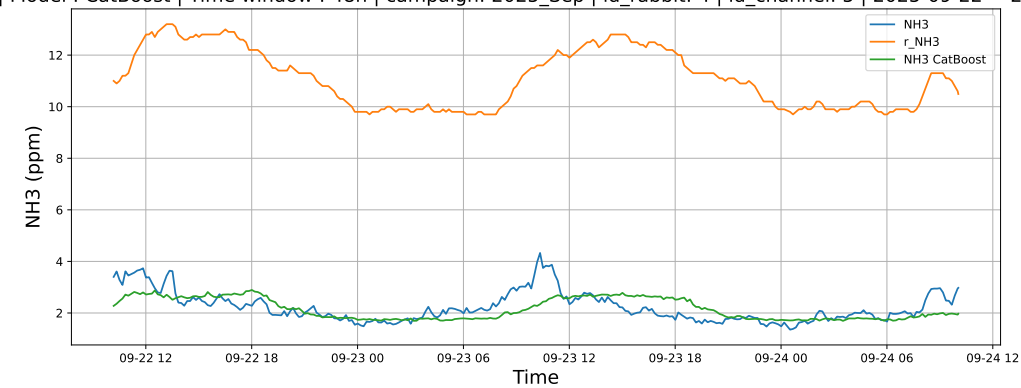
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

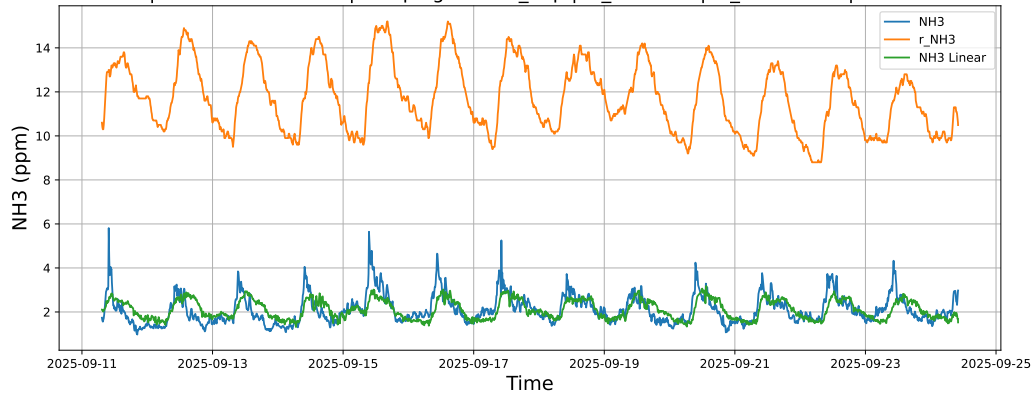


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

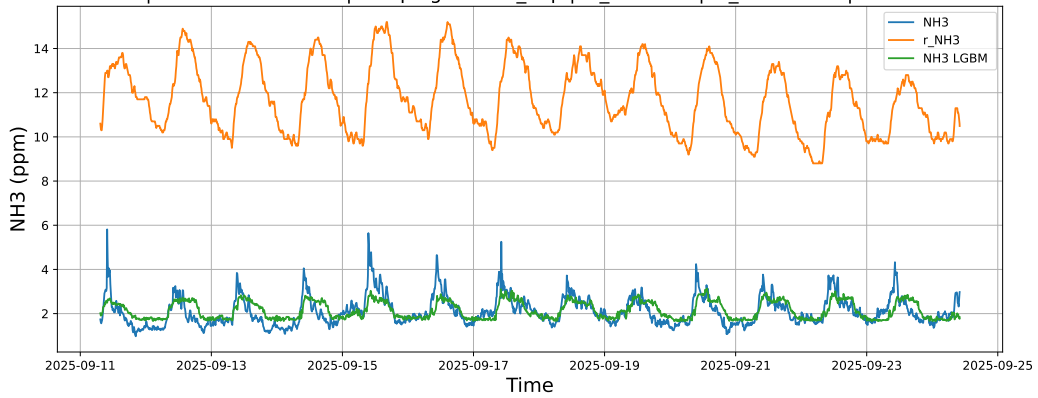


4.1.7.3 Time window : ALL

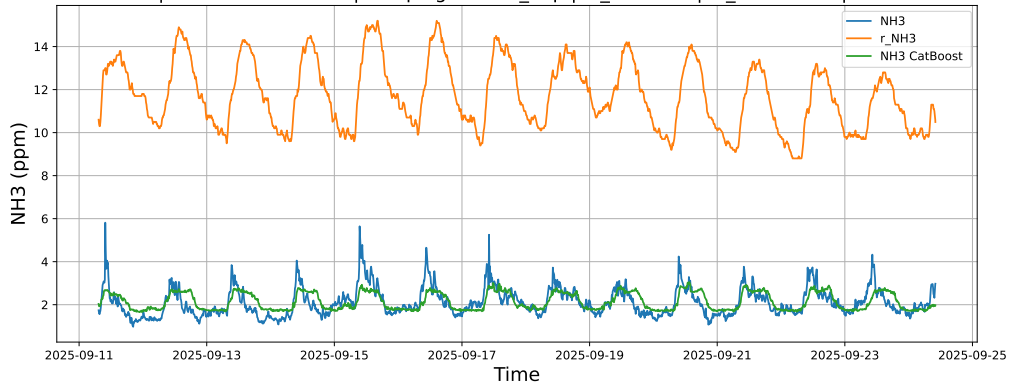
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



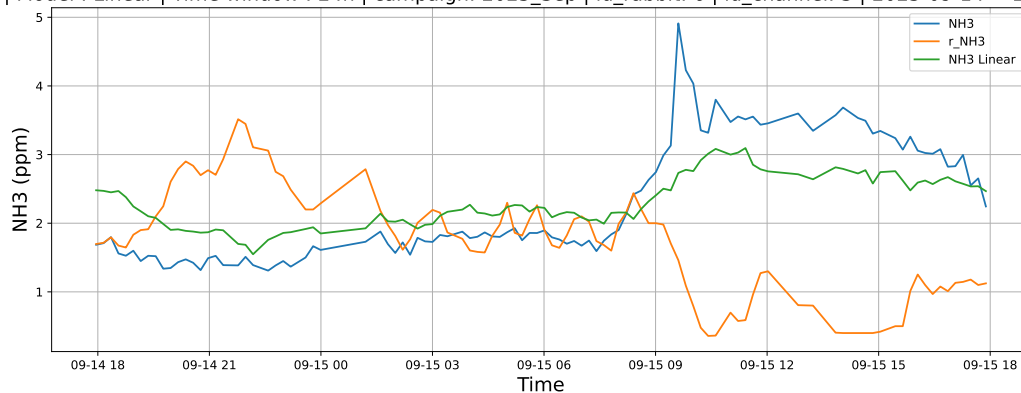
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



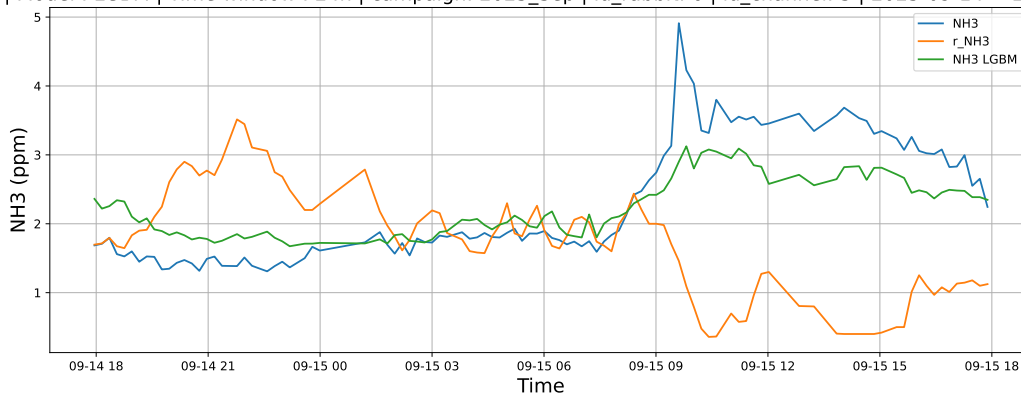
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

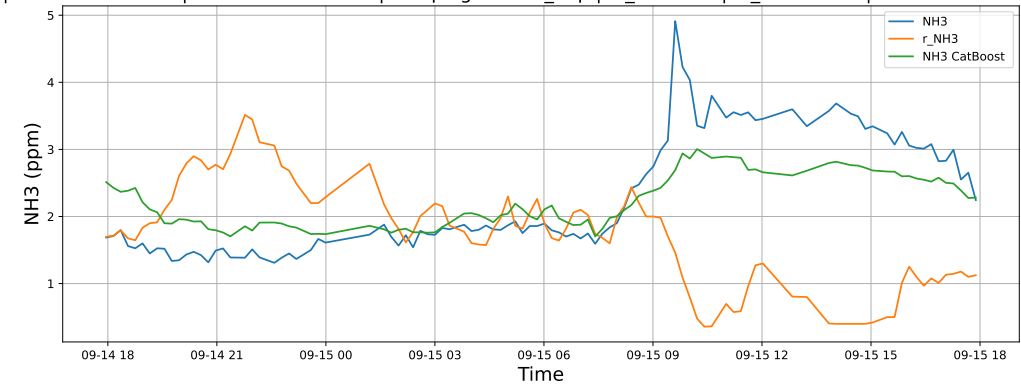
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

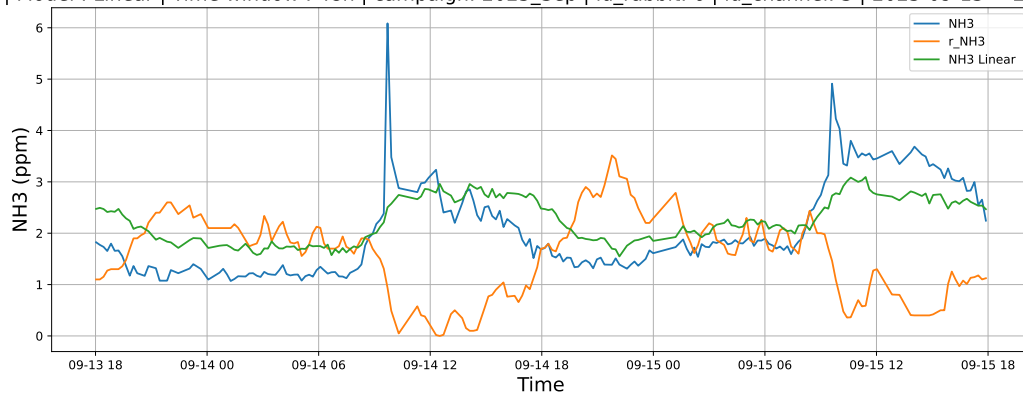


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

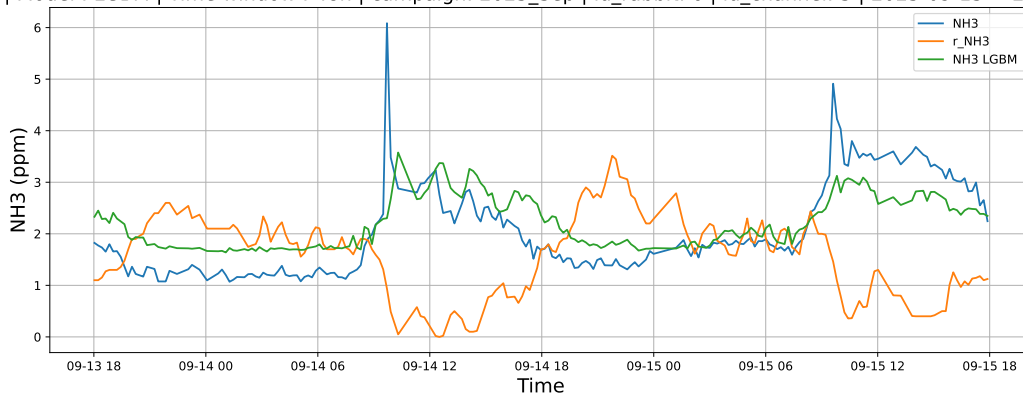


4.1.8.2 Time window : 48

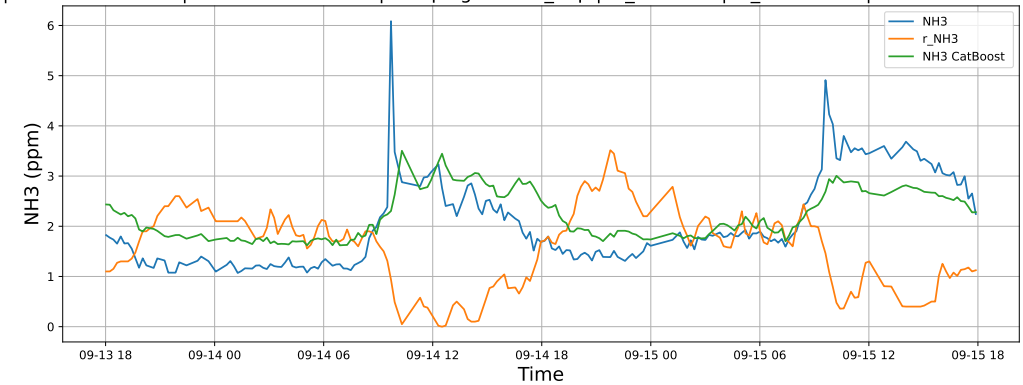
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

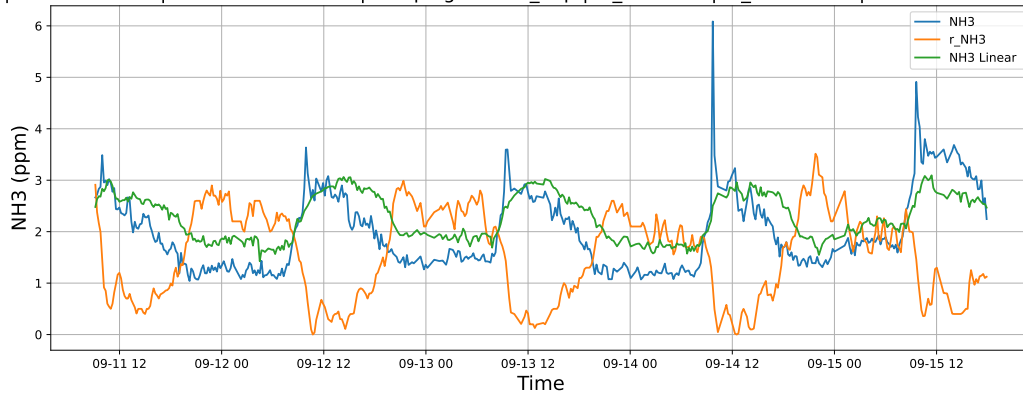


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

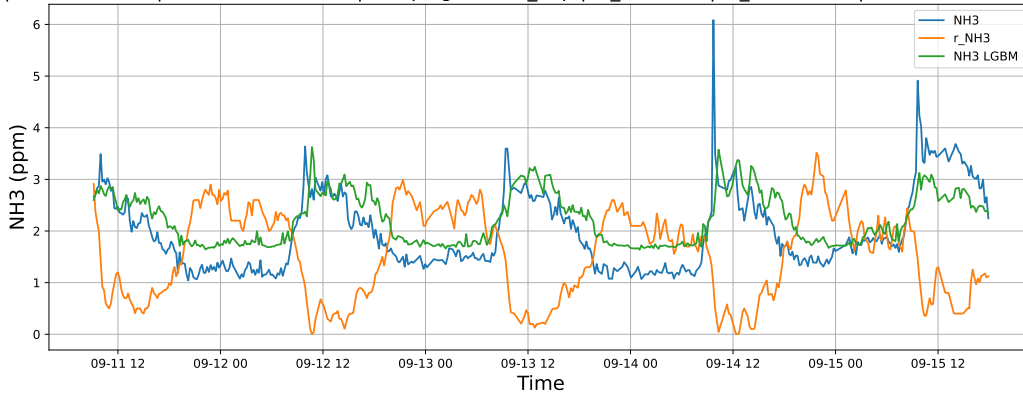


4.1.8.3 Time window : ALL

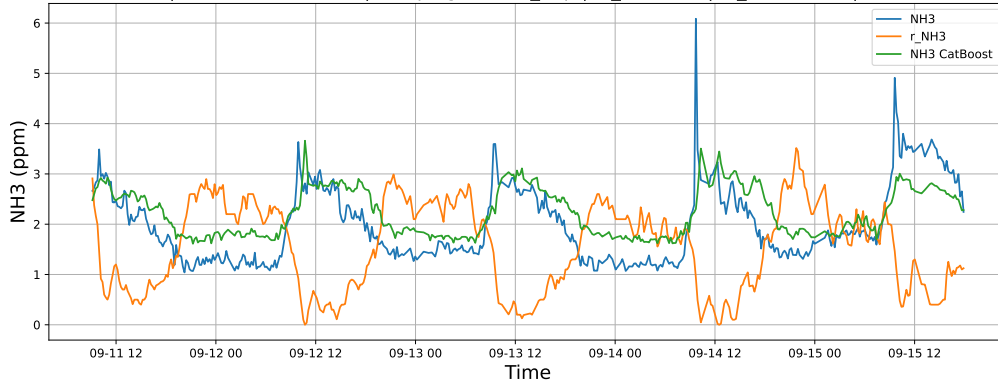
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

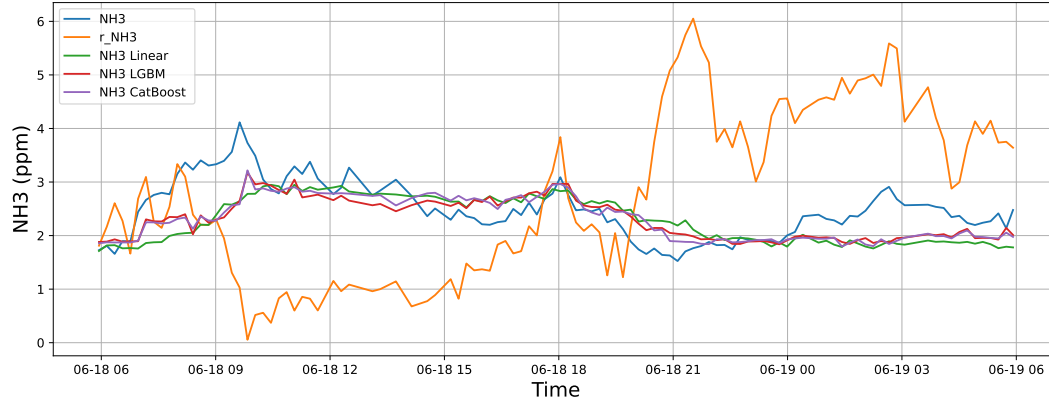


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

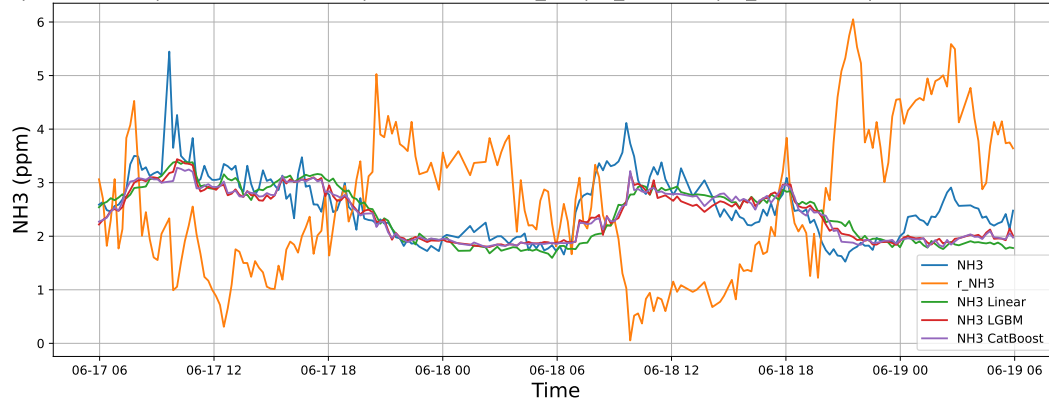
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



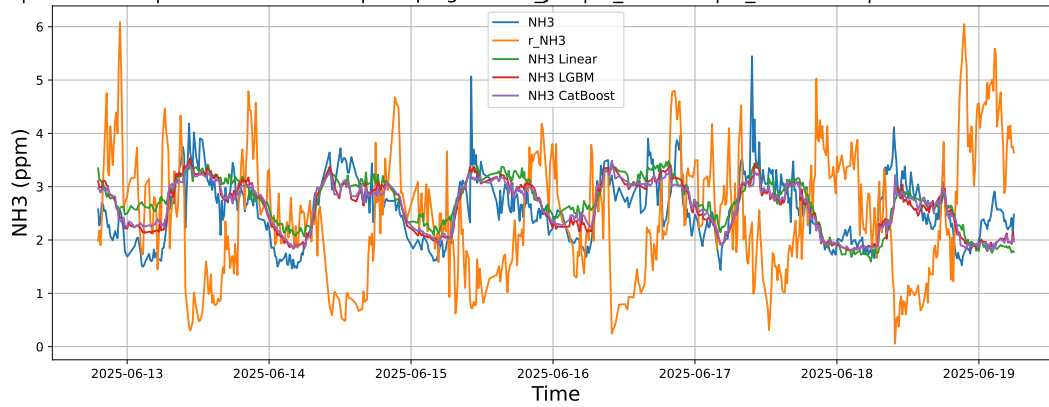
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

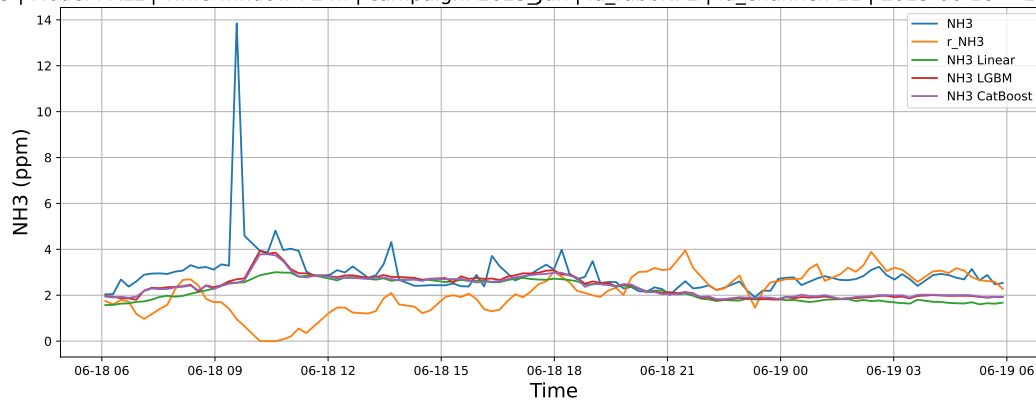
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

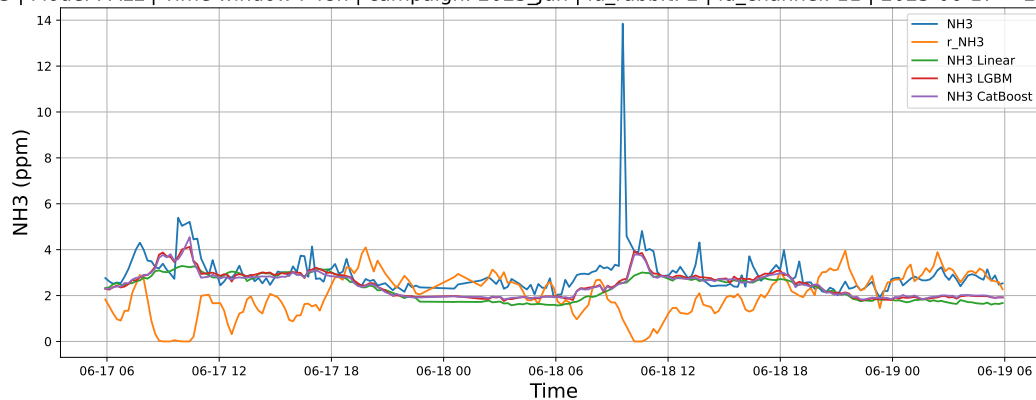
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



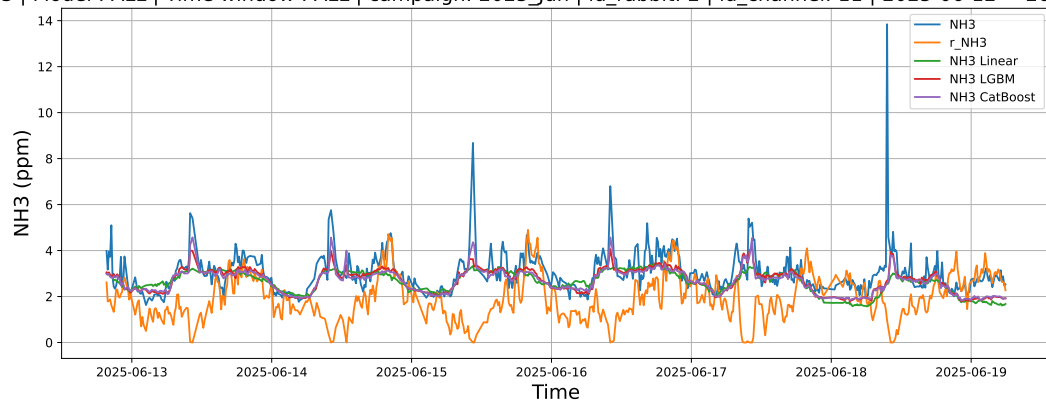
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

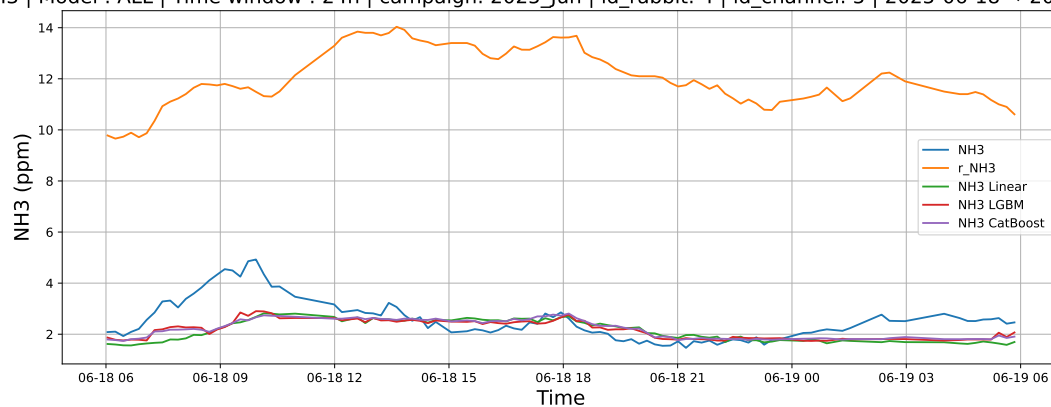
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

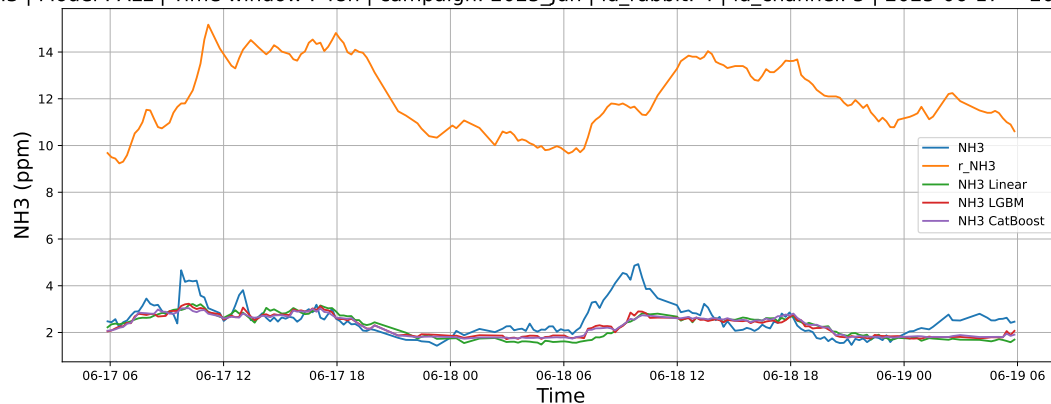
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



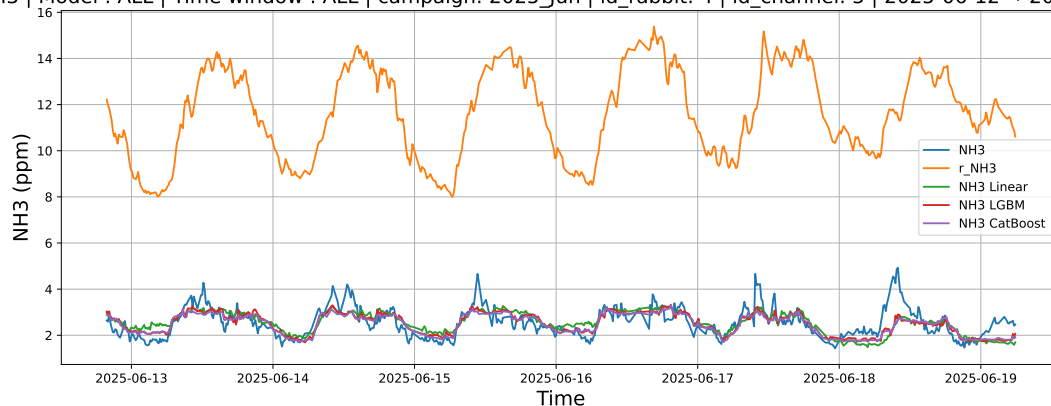
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

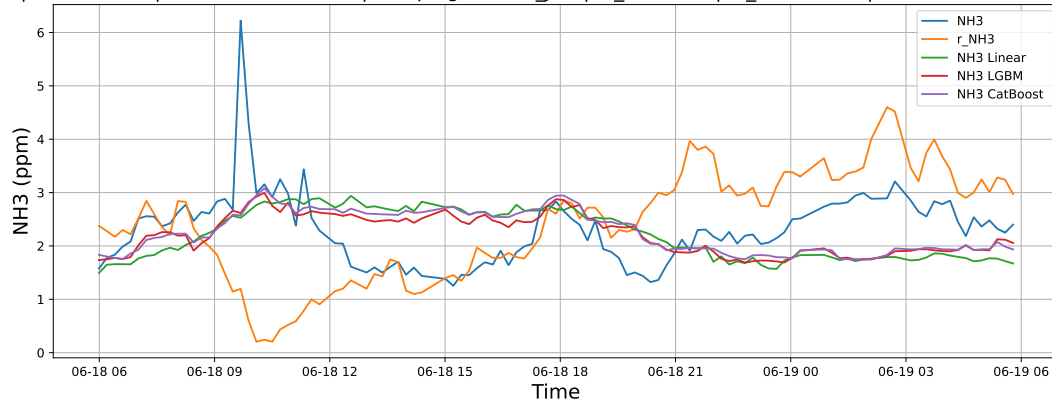
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

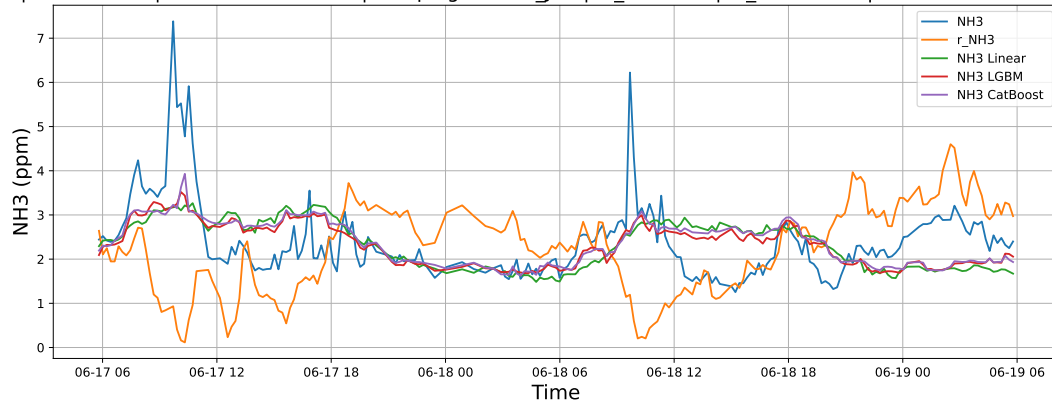
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



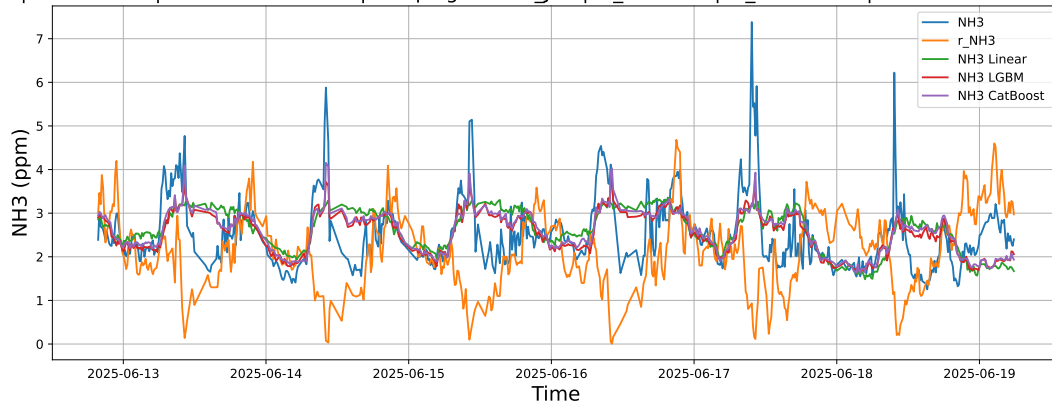
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

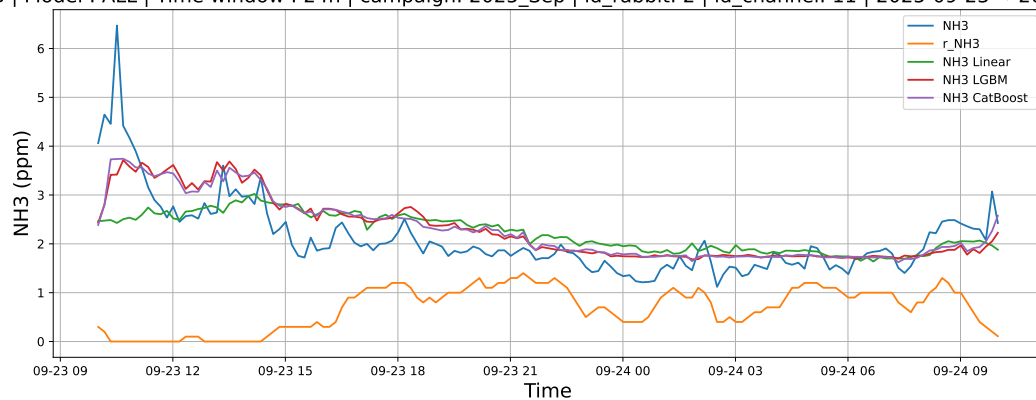
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

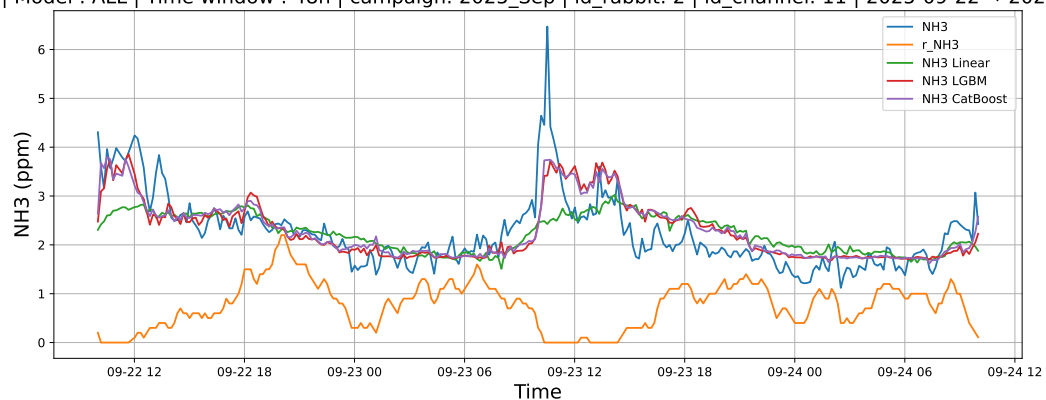
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



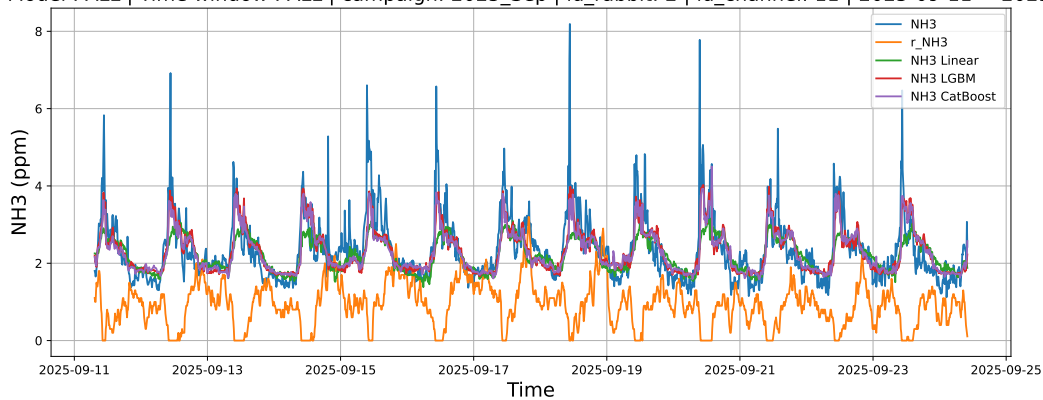
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

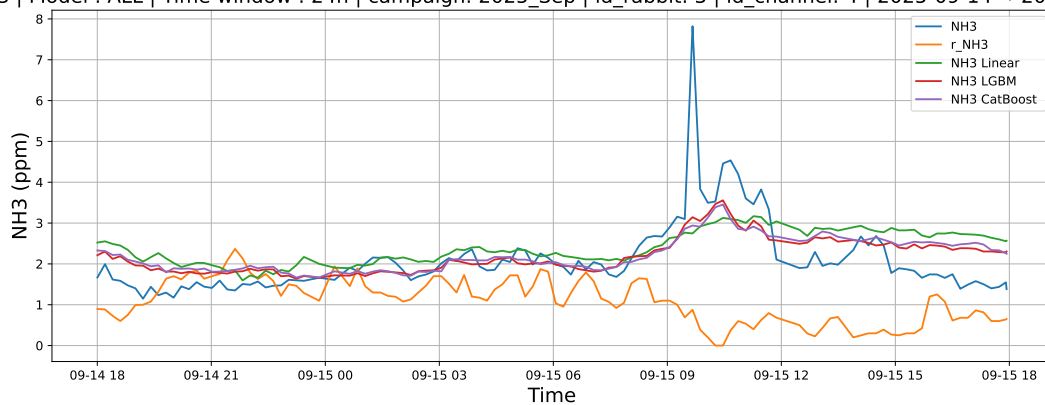
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

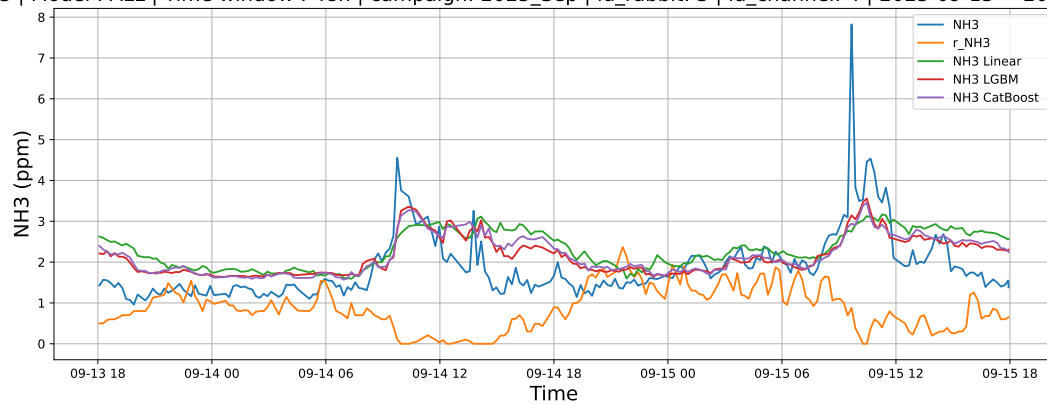
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



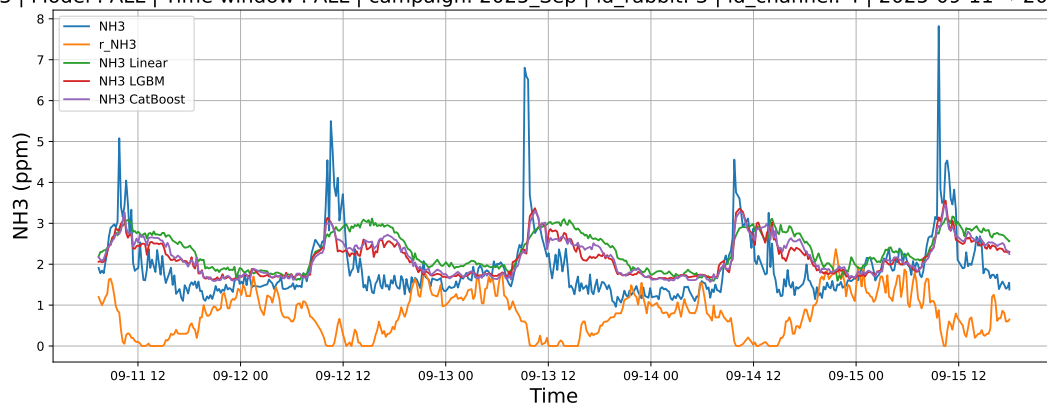
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

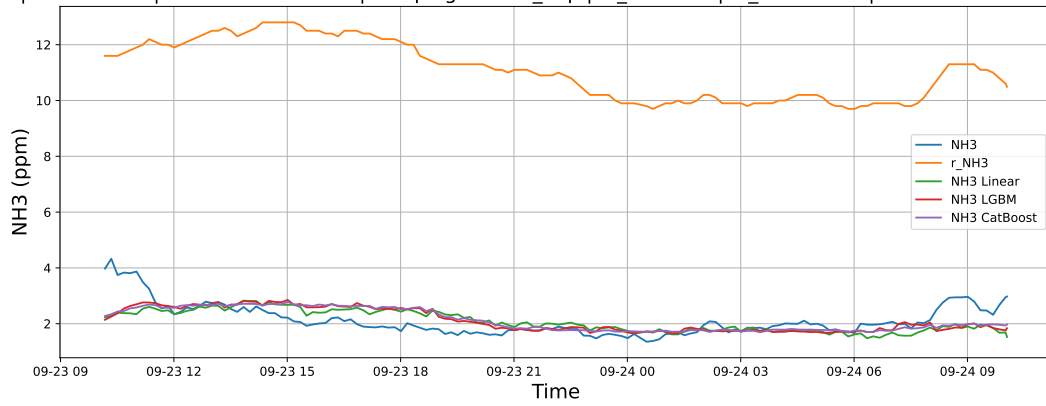
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

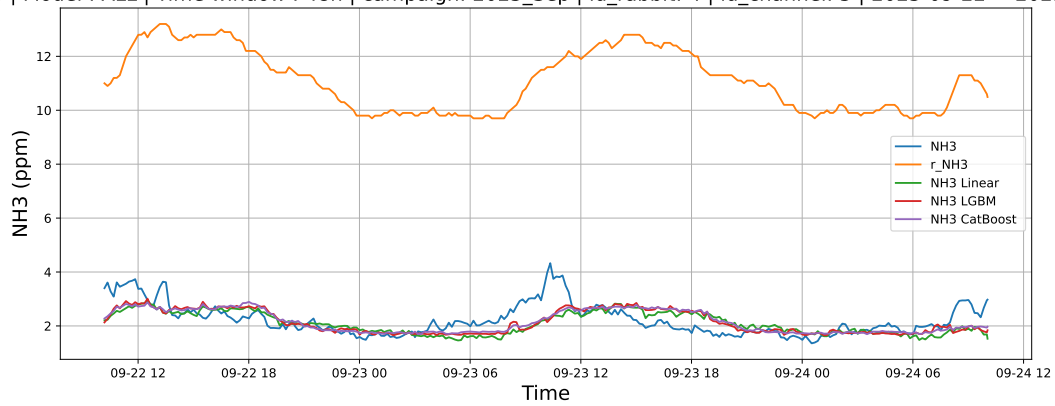
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



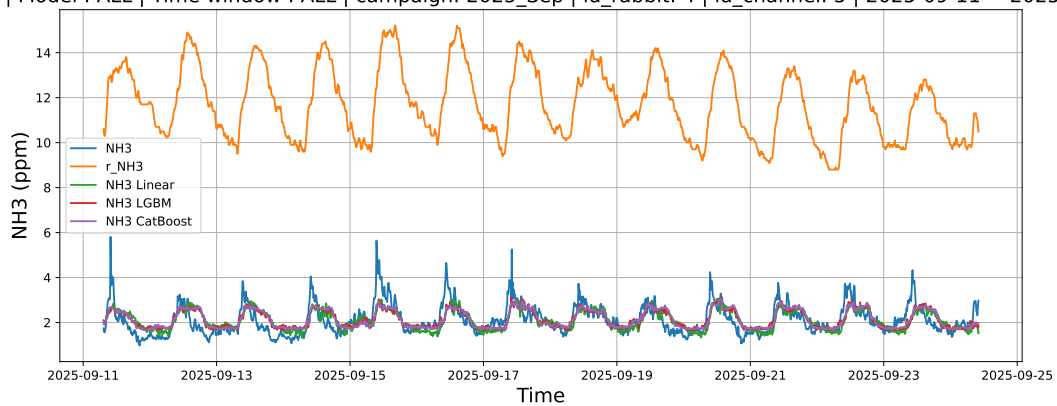
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

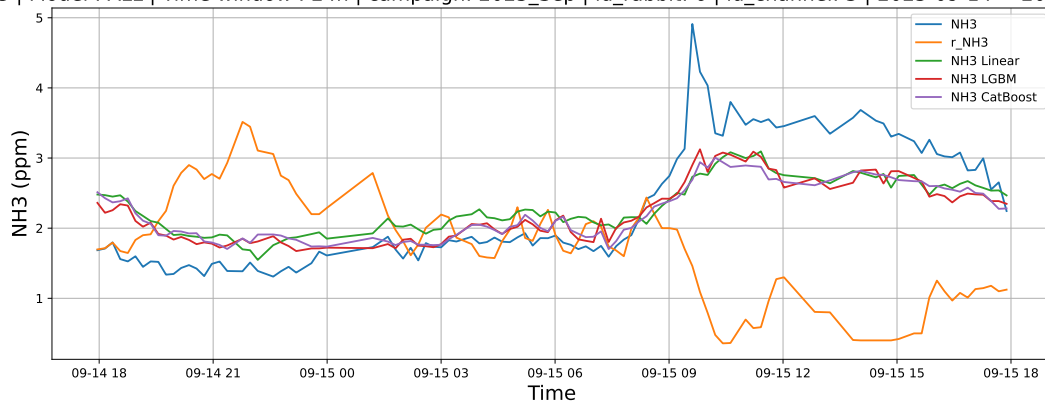
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

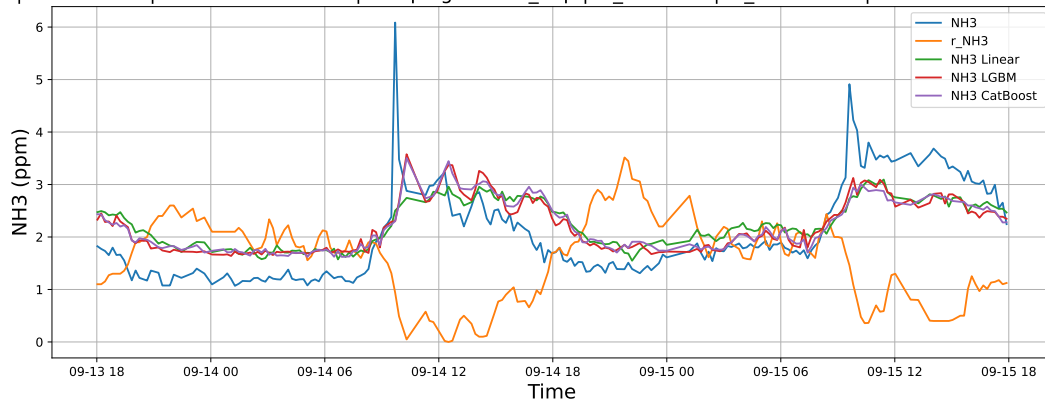
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

